Knowledge Base

Synthetic Financial Time Series Generation

From Arithmetic to GANs — Every Step Explained

Feynman Technique Throughout

Situation → Problem → Worked Arithmetic
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Chapter 1

Numbers, Algebra and Functions

Before and After

Before this chapter you need: arithmetic with whole numbers — adding, subtracting, multiplying and dividing. Nothing else.
After it you will understand: what a variable is, how to rearrange an equation without breaking it, and what people mean when they call something a *function*. Every later chapter is written in the language of functions, so this is the vocabulary the rest of the book assumes.

This chapter contains no finance and almost no notation. Its job is to make sure that when Chapter 3 writes $f(x)$ and asks how fast it changes, the symbols are already familiar rather than a second obstacle stacked on top of the idea.

1.1 Fractions, Ratios and Percentages

Situation — the everyday picture

A share costs \$40 on Monday. On Tuesday it costs \$46. You want to describe the change to someone who does not know the starting price. Saying "it went up \$6" is useless to them — \$6 on a \$40 share is a large move, \$6 on a \$4,000 share is a rounding error. What they want is the change *relative to where it started.*

Problem — where it breaks or why it matters

Absolute differences cannot be compared across things of different size. A \$6 move in Brazil and a \$6 move in China tell you nothing about which market had the bigger day. To compare, every quantity must first be put on a common scale, and the natural common scale is "fraction of the starting value".

Worked Arithmetic — concrete numbers

Start 40, end 46.

Change: $46 - 40 = 6$.
As a fraction of the start: $\dfrac{6}{40} = \dfrac{3}{20} = 0.15$.
As a percentage: $0.15 \times 100 = 15\%$.

Check it the other way round. If 40 grows by 15%, the new value is $40 \times (1 + 0.15) = 40 \times 1.15 = 46$. ✓

Now the same move on a bigger share. Start 4,000, end 4,006: $\dfrac{6}{4000} = 0.0015 = 0.15\%$. Same \$6, one hundredth of the move.

Formal Name — the mathematics

For a quantity that starts at P_0 and ends at P_1, the **simple return** is

$$R = \frac{P_1 - P_0}{P_0} = \frac{P_1}{P_0} - 1.$$

A **percentage** is a fraction multiplied by 100; the symbol % means "divide by 100". The quantity P_1/P_0 is called the **gross return** or **price relative**: a gross return of 1.15 and a simple return of 0.15 say the same thing.

Where the Analogy Breaks

"15% up" and "15% down" are not opposite moves. Going up 15% from 40 gives 46; coming back down 15% from 46 gives $46 \times 0.85 = 39.1$, not 40. The percentage is always taken relative to *wherever you currently are*, and after the first move that is a different place. This asymmetry is not a curiosity — it is the reason Chapter 2 replaces percentages with logarithms for everything in this thesis.

Why It Matters Here — link to the thesis

Every number in this thesis is a return, not a price. The five BRICS markets trade at wildly different index levels, and only by converting to returns can BOVESPA and SHANGHAI be placed on the same axis, fed to the same model, and scored by the same metric.

1.2 Variables and Rearranging Equations

Situation — the everyday picture

A recipe says: to find the cost of a taxi, take \$3 for the pickup and add \$2 for every kilometre. You could write this out in words every time, or you could write $C = 3 + 2k$ and let k stand for however many kilometres you happen to travel. The letter is shorthand for "a number I have not decided yet".

Problem — where it breaks or why it matters

Once the relationship is written down, you often need to run it backwards. The formula gives cost from distance, but the question you actually have is "I have \$15 — how far can I go?". That means solving for k instead of C, and doing that safely requires a rule for what you are allowed to do to an equation.

Worked Arithmetic — concrete numbers

The rule is: *whatever you do to one side, do to the other.* An equation is a claim that two things are equal, and equals stay equal if treated identically.

Start with $C = 3 + 2k$ and set $C = 15$:

$$15 = 3 + 2k$$

Subtract 3 from both sides:

$$15 - 3 = 3 + 2k - 3 \quad \Longrightarrow \quad 12 = 2k$$

Divide both sides by 2:

$$\frac{12}{2} = \frac{2k}{2} \quad \Longrightarrow \quad 6 = k$$

Check by substitution: $C = 3 + 2 \times 6 = 3 + 12 = 15$. ✓

Substituting the answer back into the original equation is the only check you ever need, and it costs ten seconds.

Formal Name — the mathematics

A **variable** is a symbol standing for a number that may take different values. A **constant** is a fixed number. An **equation** asserts that two expressions are equal; to **solve** it for a variable is to produce an equivalent equation with that variable alone on one side.
The operations that preserve equality are: adding or subtracting the same quantity from both sides, and multiplying or dividing both sides by the same *non-zero* quantity.

Where the Analogy Breaks

The "do the same to both sides" rule has one trap: multiplying both sides by zero. From $5 = 5$ we may multiply both sides by 0 and get $0 = 0$, which is true but has lost all information — and the step cannot be undone. This is why division by zero is forbidden: it is the reverse of a step that destroys information. The same caution reappears in Chapter 5, where a matrix with determinant zero cannot be inverted for exactly this reason.

Why It Matters Here — link to the thesis

Rearranging is how the definition of a log return, $r = \ln(P_1/P_0)$, is turned into the reconstruction $P_1 = P_0 e^r$ used whenever a synthetic return series must be turned back into a synthetic price path.

1.3 What a Function Is

Situation — the everyday picture

A vending machine has a keypad. You press A4 and a specific bar of chocolate comes out. You press A4 again tomorrow and the *same* kind of bar comes out. Press B2 and something else comes out. The machine is a rule that turns each input into exactly one output, reliably, every time.

Problem — where it breaks or why it matters

The reliability is the whole point. A machine that sometimes gave chocolate and sometimes gave crisps for the same button would be useless — you could not reason about it or plan around it. Mathematics needs the same guarantee before it can say anything about how outputs change when inputs change.

Worked Arithmetic — concrete numbers

Take the rule "double it, then add one", written $f(x) = 2x + 1$.

Input x	Working	Output $f(x)$
0	$2 \times 0 + 1$	1
1	$2 \times 1 + 1$	3
2	$2 \times 2 + 1$	5
3	$2 \times 3 + 1$	7
-1	$2 \times (-1) + 1$	-1

Each input appears once and produces one output. Note that outputs *may* repeat for different inputs without breaking anything: for $g(x) = x^2$ we get $g(2) = 4$ and $g(-2) = 4$. Two doors, same room. That is allowed. What is not allowed is one door opening onto two rooms.

Formal Name — the mathematics

A **function** f from a set A to a set B assigns to each element $x \in A$ exactly one element $f(x) \in B$. We write $f : A \to B$, read "f maps A to B". The symbol $\in$ means "is an element of".

The value $f(x)$ is the **image** of x. The notation is deliberately mechanical: f is the machine, x is what goes in, $f(x)$ is what comes out.

Where the Analogy Breaks

The vending-machine picture suggests the machine is simple and you could open it up and see the mechanism. Many functions in this thesis are not like that. A trained neural network is a function — one input gives one output, reliably — but it has hundreds of thousands of internal numbers and no human-readable mechanism. "Function" guarantees only the input–output discipline, never simplicity or interpretability.

Why It Matters Here — link to the thesis

A generator in a GAN is a function: it takes a vector of random noise and returns a synthetic return series. A discriminator is a function: it takes a series and returns a score. Training changes *which* function these are, but they are functions at every moment, which is what makes it meaningful to ask how their outputs respond to their inputs.

1.4 Domain and Range

Situation — the everyday picture

The vending machine has buttons A1 to D5 and nothing else. Pressing a letter that is not on the keypad does nothing — the question "what does the machine give for Z9?" has no answer, because Z9 was never a legal input.

Problem — where it breaks or why it matters

Ignoring which inputs are legal produces confident nonsense. A formula that divides by x is silent about what happens at $x = 0$ until someone asks; a formula involving $\sqrt{x}$ has nothing to say about $x = -1$ if we are working with ordinary numbers. Stating the legal inputs up front prevents the error.

Worked Arithmetic — concrete numbers

Example 1: $f(x) = 1/x$.
Legal inputs: every number except 0. At $x = 2$, $f = 0.5$; at $x = 0.1$, $f = 10$; at $x = 0.001$, $f = 1000$. As x shrinks toward zero the output grows without bound, and at $x = 0$ there is no output at all.

Example 2: $f(x) = \sqrt{x}$ over ordinary (real) numbers.
Legal inputs: $x \geq 0$. Outputs: $\sqrt{0} = 0$, $\sqrt{9} = 3$, $\sqrt{2} \approx 1.414$. Every output is ≥ 0, so although the legal inputs run from 0 upwards, so do the outputs.

Example 3: $f(x) = x^2$.
Legal inputs: all numbers. Outputs: never negative, because a number times itself is never negative. So the outputs occupy only half the number line even though the inputs occupy all of it.

Formal Name — the mathematics

The **domain** of f is the set of inputs on which f is defined. The **codomain** is the set the outputs are declared to live in. The **range** (or image) is the set of values f actually attains. For $f(x) = x^2$ with domain $\mathbb{R}$ (all real numbers), the codomain may be declared as $\mathbb{R}$, but the range is only $[0, \infty)$ — the non-negative numbers. Range is a subset of codomain, sometimes a strict one.

Where the Analogy Breaks

Domain is often presented as a technicality to be recited and forgotten. It is not. In Chapter 2 the domain of ln is exactly the strictly positive numbers, and that single restriction is what forces prices — never returns — to be the thing we take logarithms of. The restriction carries real content about what the mathematics is willing to model.

Why It Matters Here — link to the thesis

The generators in this thesis end with a tanh activation, whose range is $(-1, 1)$. That is a deliberate choice of range: it guarantees the network cannot emit an arbitrarily large value, and it is why the training data must first be squashed into the same interval before the two can be compared.

1.5 Linear Functions and Slope

Situation — the everyday picture

You are walking up a straight ramp. For every 4 metres you move forward, you rise 1 metre. The ramp does not get steeper or shallower — the trade-off between forward and upward is the same everywhere on it. One number, 1/4, describes the entire ramp.

Problem — where it breaks or why it matters

Most relationships are not ramps. But a great deal of mathematics consists of finding the ramp that best matches a curve near a point of interest, because ramps are the only shape we can reason about effortlessly. Before that programme can start, "steepness" has to be pinned to a number.

Worked Arithmetic — concrete numbers

Take $f(x) = 3x + 2$.

x	$f(x)$	Change in x	Change in f
1	5	—	—
2	8	1	3
4	14	2	6
7	23	3	9

From $x = 1$ to $x = 4$: heights 5 and 14, change 9, horizontal distance 3, ratio $9/3 = 3$.
From $x = 4$ to $x = 7$: heights 14 and 23, change 9, distance 3, ratio 3.

The ratio is 3 no matter which two points we pick. That constancy is precisely what makes the function linear, and the 3 is exactly the number multiplying x in the formula. The $+2$ shifts the whole line up without tilting it: at $x = 0$ the value is 2.

Formal Name — the mathematics

A **linear function** (strictly, an *affine* one) has the form

$$f(x) = mx + c,$$

where m is the **slope** or **gradient** and c is the **intercept**, the value at $x = 0$. For any two distinct points (x_1, y_1) and (x_2, y_2) on the line,

$$m = \frac{y_2 - y_1}{x_2 - x_1},$$

often read "rise over run". Positive m tilts upward, negative m downward, and $m = 0$ is flat.

Where the Analogy Breaks

Slope as "rise over run" silently assumes the line is not vertical. A vertical line has zero run, and the ratio would require dividing by zero — so vertical lines simply have no slope, rather than an infinite one. Refusing to write $m = \infty$ here is the same discipline that will matter in Chapter 9, where a distribution with an undefined fourth moment is genuinely undefined rather than merely large.

Why It Matters Here — link to the thesis

The derivative in Chapter 3 is nothing more than this slope, computed over a vanishingly small run. Gradient descent — the algorithm that trains every model in this thesis — reads that slope and steps downhill.

1.6 Quadratics and Curvature

Situation — the everyday picture

Throw a ball straight up. It rises, slows, stops, and falls. Its height against time is not a ramp: the steepness changes continuously, from strongly positive at the throw to zero at the top to strongly negative on the way down.

Problem — where it breaks or why it matters

A single slope cannot describe this path, because the path has a different slope at every instant. We need a shape with built-in bend, and the simplest such shape is the one produced by squaring.

Worked Arithmetic — concrete numbers

Take $f(x) = x^2$.

x	$f(x)$	Interval	Change in f	Average slope
0	0	—	—	—
1	1	$0 \to 1$	1	1
2	4	$1 \to 2$	3	3
3	9	$2 \to 3$	5	5
4	16	$3 \to 4$	7	7

The average slope over each unit step is $1, 3, 5, 7$ — rising by 2 every time. The function is not merely increasing; the *rate* at which it increases is itself increasing at a steady rate. That steady second-level change is what "curvature" means, and here it equals 2.

Note also the symmetry: $f(-3) = 9 = f(3)$. The curve is a mirror image about the vertical line $x = 0$, which sits at the bottom of the bowl.

Formal Name — the mathematics

A **quadratic function** has the form

$$f(x) = ax^2 + bx + c, \qquad a \neq 0.$$

Its graph is a **parabola**. If $a > 0$ it opens upward and has a single lowest point; if $a < 0$ it opens downward and has a single highest point. That turning point sits at $x = -b/(2a)$. For $f(x) = x^2$ we have $a = 1$, $b = 0$, $c = 0$, so the turning point is at $x = 0$, matching the table above.

Where the Analogy Breaks

A parabola has exactly one turning point, which makes "find the lowest value" easy: there is nowhere else it could be. Almost none of the functions minimised in this thesis are like that. A neural network's loss surface has enormous numbers of local dips, and an algorithm that walks downhill will stop at whichever one it happens to reach. The single-minimum picture is the exception, not the rule — Chapter 4 returns to this.

Why It Matters Here — link to the thesis

Squared error — the quantity minimised when the TimeGAN autoencoder learns to reconstruct its input — is a quadratic in the reconstruction error. Its bowl shape is why the reconstruction loss has a well-defined best answer, and why a reported R^2 of +0.995 means the bottom of that bowl was very nearly reached.

1.7 Composition of Functions

Situation — the everyday picture

A factory line has two stations. The first paints the part; the second boxes it. A raw part enters, a painted part passes between them, a boxed painted part comes out. Nobody inspects the middle stage — from outside, the line is one machine turning raw parts into finished goods.

Problem — where it breaks or why it matters

Deep networks are precisely this: dozens of simple stations in a row. To train them we must answer "if I adjust station one slightly, how much does the final output move?" — a question about the whole line, asked at a single station. Answering it requires first being clear about what stacking functions even means.

Worked Arithmetic — concrete numbers

Let $g(x) = x + 3$ (the painter) and $f(u) = 2u$ (the boxer). Apply g first, then f.

Input x	After g: $x + 3$	After f: $2(x + 3)$
1	4	8
2	5	10
5	8	16

The combined rule is $f(g(x)) = 2(x + 3) = 2x + 6$. Check at $x = 5$: $2 \times 5 + 6 = 16$. ✓

Order matters. Reverse the stations — box first, then paint: $g(f(x)) = 2x + 3$. At $x = 5$ that gives 13, not 16. Painting a boxed part is not the same as boxing a painted one.

Formal Name — the mathematics

Given $g : A \to B$ and $f : B \to C$, the **composition** $f \circ g$ is the function $A \to C$ defined by

$$(f \circ g)(x) = f(g(x)).$$

It is read "f after g": the function written on the *left* is applied *last*. Composition requires the range of g to lie inside the domain of f — the parts leaving the first station must be things the second station can accept.

Where the Analogy Breaks

The factory picture suggests each station is independent, so you could swap or remove one freely. In a trained network that is false. Each layer's weights have been fitted assuming exactly what the previous layer sends it; extract a middle layer and it produces nothing meaningful on raw input. The stations are separable in notation only.

Why It Matters Here — link to the thesis

A neural network is one long composition, and the chain rule of Chapter 4 is the tool that differentiates it. Everything called "backpropagation" is the chain rule applied to a composition of this kind. The TimeGAN recovery function composed with its embedder is a composition whose round-trip fidelity we measure directly — and require to be accurate to within 10^{-3} before training is allowed to proceed.

Chapter 2

Exponentials and Logarithms

Before and After

Before this chapter you need: percentages and gross returns (§1.1), functions and their domains (§1.3–§1.4), composition (§1.7).
After it you will understand: why every quantity in this thesis is a *log* return rather than a percentage return, why that choice makes multi-day returns add up exactly, and why a log-scaled histogram is the honest way to look at the tail of a distribution.

This chapter earns its length. The decision to work in log returns propagates into every model, every metric and every figure in the project, and the reasons are worth seeing in full rather than accepting on authority.

2.1 Compound Growth

Situation — the everyday picture

You put \$100 in an account paying 8% a year, and you leave the interest in. After one year you have \$108. The second year's 8% is charged on \$108, not on the original \$100, so you gain \$8.64 rather than \$8. Growth feeds on itself.

Problem — where it breaks or why it matters

Because each year's gain depends on the running total, growth over n years is not n separate additions. It is n multiplications, and multiplication is far harder to reason about than addition — especially when you want to compare periods of different length, or split a long period into pieces.

Worked Arithmetic — concrete numbers

\$100 at 8% per year, interest retained.

Year	Start	Gain	End
1	100.00	8.00	108.00
2	108.00	8.64	116.64
3	116.64	9.33	125.97

As repeated multiplication: $100 \times 1.08 \times 1.08 \times 1.08 = 100 \times 1.08^3 = 125.97$. ✓

Compare to simple (non-compounding) interest: $100 + 3 \times 8 = 124$. The gap of \$1.97 after only three years is the compounding effect, and it widens with time.

Now compound more often. Take 100% per year, but split it into n instalments of $1/n$ each:

$$\begin{aligned} n = 1: &\quad (1 + 1/1)^1 = 2.00000 \\ n = 2: &\quad (1 + 1/2)^2 = 1.5^2 = 2.25000 \\ n = 4: &\quad (1 + 1/4)^4 = 1.25^4 = 2.44141 \\ n = 12: &\quad (1 + 1/12)^{12} = 2.61304 \\ n = 365: &\quad (1 + 1/365)^{365} = 2.71457 \end{aligned}$$

The numbers climb, but ever more slowly, and they are converging on something just above 2.718. They do not run away to infinity.

Formal Name — the mathematics

An **exponential function** has the form $f(x) = a^x$ with $a > 0$. Growth at rate R per period, compounded over n periods, multiplies the starting value by $(1 + R)^n$.
The limit reached by the table above defines **Euler's number**:

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n \approx 2.718281828\ldots$$

It is irrational: its decimal expansion never terminates or repeats. The function $f(x) = e^x$ is called *the* exponential function, written $\exp(x)$.

Where the Analogy Breaks

The bank analogy suggests e arises from a peculiar accounting convention, as if it were an artefact of how often interest is credited. It is not. The quantity e^x is singled out by a property with nothing to do with money: it is the only function that is its own rate of change, a fact Chapter 3 makes precise. Continuous compounding is one place e shows up, not the reason it exists.

Why It Matters Here — link to the thesis

Continuous compounding is the natural setting for asset prices, which are quoted continuously through the trading day rather than at fixed interest dates. Writing $P_1 = P_0 e^r$ instead of $P_1 = P_0(1 + R)$ is what makes r additive across time, which is the property the rest of this thesis is built on.

2.2 The Logarithm as the Inverse

Situation — the everyday picture

A machine doubles whatever you feed it: put in 3, get 6; put in 5, get 10. Now someone hands you a 32 and asks what was fed in. You need the machine run backwards. Since $2^5 = 32$, the answer is 5 — you are asking "how many doublings?".

Problem — where it breaks or why it matters

Running exponentials backwards is exactly the question “what power was this?”. It cannot be answered by any combination of the four arithmetic operations, so it needs a name and a function of its own. Without one, every compound-growth calculation is a one-way street.

Worked Arithmetic — concrete numbers

Base 2, so we are counting doublings.

x	2^x	$\log_2$ of it
0	1	$\log_2 1 = 0$
1	2	$\log_2 2 = 1$
2	4	$\log_2 4 = 2$
3	8	$\log_2 8 = 3$
5	32	$\log_2 32 = 5$

Read the table left to right for exponentials, right to left for logarithms. The two columns hold identical information.

The characteristic move. Note that $4 \times 8 = 32$, and in the log column $2 + 3 = 5$. Multiplying the numbers corresponds to adding their logarithms. This is not a coincidence of these particular values; it is the defining property, and it is the entire reason logarithms are useful.

With base e. The natural logarithm ln undoes e^x: $\ln(1) = 0$, $\ln(e) = 1$, $\ln(e^2) = 2$. Numerically $\ln(2) \approx 0.693$ and $\ln(10) \approx 2.303$.

Formal Name — the mathematics

The **logarithm to base a**, written $\log_a$, is the inverse of $x \mapsto a^x$:

$$\log_a(y) = x \iff a^x = y.$$

The **natural logarithm** $\ln = \log_e$ satisfies

$$e^{\ln x} = x \quad \text{and} \quad \ln(e^x) = x.$$

Its domain is the strictly positive reals: $\ln(0)$ and $\ln(-1)$ are undefined, because no real power of e produces zero or a negative number. As $x \to 0^+$, $\ln x \to -\infty$.

Where the Analogy Breaks

Calling ln “the inverse” invites the assumption that it undoes e^x in every sense, including numerically. It does not. Computing e^{-40} gives roughly 4×10^{-18}, which in ordinary floating-point arithmetic may round to something whose logarithm is no longer -40. The mathematical inverse is exact; the computational one is not, which is why numerical code so often carries quantities in log form from the start rather than converting back and forth.

Why It Matters Here — link to the thesis

The strictly-positive domain is the reason we take logarithms of *prices* and never of returns. A price is always positive; a return is negative on roughly half of all days, and ln has nothing to say about those.

2.3 The Log Rules

Situation — the everyday picture

Before electronic calculators, engineers multiplied large numbers by looking up two logarithms in a printed table, adding them by hand, and looking the result back up. A hard operation was traded for an easy one at the cost of two lookups.

Problem — where it breaks or why it matters

The same trade is worth making today, for a different reason. Products of many terms overflow or underflow the range a computer can represent, and they resist algebraic manipulation. Sums do neither. Anything that converts products to sums buys both numerical stability and analytical tractability.

Worked Arithmetic — concrete numbers

Verify each rule on numbers small enough to check mentally, using base 2.

Product rule. $\log_2(4 \times 8) = \log_2 32 = 5$, and $\log_2 4 + \log_2 8 = 2 + 3 = 5$. ✓

Quotient rule. $\log_2(32/4) = \log_2 8 = 3$, and $\log_2 32 - \log_2 4 = 5 - 2 = 3$. ✓

Power rule. $\log_2(4^3) = \log_2 64 = 6$, and $3 \times \log_2 4 = 3 \times 2 = 6$. ✓

Log of 1. $\log_2 1 = 0$, because $2^0 = 1$. This holds in every base, and it is the fact that makes "no change" correspond to a log return of exactly zero.

Formal Name — the mathematics

For $x, y > 0$ and any real k:

$$\ln(xy) = \ln x + \ln y, \qquad \ln\left(\frac{x}{y}\right) = \ln x - \ln y, \qquad \ln(x^k) = k \ln x, \qquad \ln 1 = 0.$$

To change base: $\log_a x = \ln x / \ln a$.

Where the Analogy Breaks

There is no rule for $\ln(x + y)$. The logarithm converts multiplication into addition, not addition into anything simpler — and beginners routinely invent $\ln(x + y) = \ln x + \ln y$, which is false. Check it: $\ln(2 + 2) = \ln 4 \approx 1.386$, while $\ln 2 + \ln 2 \approx 1.386$. Those agree, which is exactly the sort of accident that entrenches the error. Try $\ln(1 + 3) = \ln 4 \approx 1.386$ against $\ln 1 + \ln 3 = 0 + 1.099 = 1.099$. They differ.

Why It Matters Here — link to the thesis

The product rule is the engine of the next section. It is also why the log-likelihood, rather than the likelihood, is what gets maximised whenever a GARCH model is fitted to a return series in this project: a product of thousands of small probabilities underflows to zero, while the corresponding sum of logarithms is perfectly well behaved.

2.4 Why Log Returns Are Additive

Situation — the everyday picture

You invest \$100. It rises 10%, then falls 10%. You are back where you started — or so the symmetry of the two numbers suggests. You are not. You have \$99.

Problem — where it breaks or why it matters

Percentage returns do not add across time. Over two days the errors are small enough to ignore; over 5,000 trading days — the span of this project's data — they compound into nonsense. Any quantity that is to be summed, averaged, or modelled as a sequence of increments must be additive by construction.

Worked Arithmetic — concrete numbers

The asymmetry, in full. Start at 100.

$$100 \times 1.10 = 110, \qquad 110 \times 0.90 = 99.$$

Net change -1%, not 0%. In gross terms $1.10 \times 0.90 = 0.99$.

Now in logs:

$$\ln(1.10) = +0.09531, \qquad \ln(0.90) = -0.10536.$$

Sum: $+0.09531 - 0.10536 = -0.01005$. And directly, $\ln(99/100) = \ln(0.99) = -0.01005$. ✓
The log returns sum to exactly the right answer. The percentages do not.

The genuinely symmetric case. Take a rise and fall that really do cancel: up by a factor 1.10, then down by a factor $1/1.10 = 0.909091$.

$$\ln(1.10) = +0.09531, \qquad \ln(0.909091) = -0.09531.$$

They sum to exactly zero, and indeed $100 \times 1.10 \times 0.909091 = 100$. In log space, "up then back down" is $+a$ then $-a$ — perfectly symmetric. In percentage space it is $+10\%$ then -9.09%, which looks lopsided but is not.

Three days from the data-shaped example. Prices 4,000, 4,080, 4,050.

$$r_1 = \ln(4080/4000) = \ln(1.02) = +0.019803$$

$$r_2 = \ln(4050/4080) = \ln(0.992647) = -0.007380$$

Sum: $+0.012423$. Direct: $\ln(4050/4000) = \ln(1.0125) = +0.012423$. ✓
By percentages: $+2.000\% - 0.735\% = +1.265\%$, whereas the true change is $+1.250\%$. An error of 1.5 basis points over two days — and it accumulates.

Formal Name — the mathematics

The **log return** on day t is

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln P_t - \ln P_{t-1}.$$

Summing over T days telescopes:

$$\sum_{t=1}^{T} r_t = (\ln P_1 - \ln P_0) + (\ln P_2 - \ln P_1) + \cdots + (\ln P_T - \ln P_{T-1}) = \ln P_T - \ln P_0 = \ln\left(\frac{P_T}{P_0}\right),$$

every interior term cancelling with its neighbour. The relationship to the simple return R_t is $r_t = \ln(1 + R_t)$, and for small moves $r_t \approx R_t$: at $R = 0.02$ we get $r = 0.019803$, agreeing to three decimal places.

Where the Analogy Breaks

Log returns are additive across *time* but not across *assets*. The log return of a portfolio is not the weighted sum of its constituents' log returns, because a sum of prices does not decompose under a logarithm — there is no rule for $\ln(x + y)$, as §2.3 insisted. Portfolio work therefore often returns to simple returns, which *are* additive across assets and not across time. Each convention buys one form of additivity and forfeits the other.

Why It Matters Here — link to the thesis

Every model, metric and figure in this thesis consumes r_t. Prices are never used directly: they trend, they live on incomparable scales across the five BRICS markets, and they violate the stationarity that Chapter 8 requires. Converting to log returns fixes all three problems in one step.

2.5 Reading Tails on a Log Scale

Situation — the everyday picture

A histogram of daily returns is a spike at zero with two barely visible smears at the edges. The bar at zero is thousands of days tall; the bar at −8% holds three days. Drawn on ordinary axes, the three-day bar is invisible — less than one pixel — and the picture says only "returns are usually small", which everybody knew.

Problem — where it breaks or why it matters

The rare days are the ones that matter. A risk model is judged almost entirely on how it handles the far edges of the distribution, and those edges are exactly what a linear axis erases. We need a way of drawing the picture in which a count of 3 and a count of 3,000 are both legible.

Worked Arithmetic — concrete numbers

Suppose four bins hold counts 3,000, 300, 30, 3.

On a linear axis, with the tallest bar 10 cm high, the others are 1 cm, 1 mm and 0.1 mm. The last is thinner than the ink.

On a log axis, we plot $\log_{10}$ of each count:

$$\log_{10} 3000 = 3.477, \quad \log_{10} 300 = 2.477, \quad \log_{10} 30 = 1.477, \quad \log_{10} 3 = 0.477.$$

The heights are now $3.477, 2.477, 1.477, 0.477$ — equally spaced, one unit apart. Every bar is visible, and the constant gap of exactly 1 reveals that each bin holds one tenth of the previous, a tenfold decay that the linear plot concealed completely.

Formal Name — the mathematics

A **logarithmic axis** places a value v at position $\log v$. Equal distances then represent equal *ratios* rather than equal differences: the gap from 1 to 10 is the same length as from 10 to 100.
A distribution whose tail decays as a power law, $\mathbb{P}(|X| > x) \sim x^{-\alpha}$, becomes a straight line of slope $-\alpha$ when both axes are logarithmic, since

$$\ln \mathbb{P}(|X| > x) = -\alpha \ln x + \text{constant}.$$

Straightness on a log–log plot is the visual signature of a power law, and the slope reads off the tail index directly.

Where the Analogy Breaks

A log axis cannot display a count of zero, since $\ln 0$ is undefined — empty bins simply vanish rather than sitting on the axis. It also flatters the eye: differences that are genuinely enormous are compressed into a modest vertical gap, so a model that is off by a factor of ten can look close. The log axis makes the tail *visible*; it does not make disagreements there *small*.

Why It Matters Here — link to the thesis

The stylised-facts figure produced for every market in this project includes a log-scale distribution panel for exactly this reason, with values outside the plotted range clipped rather than dropped so that no observation silently disappears. The panel is where a generator that has matched the centre of the distribution but missed its tails is caught — and Chapter 9 shows that missing the tails is the characteristic failure of these models.

Chapter 3

Calculus I — Change

Before and After

Before this chapter you need: functions, domain and range (§1.3–§1.4), slope of a straight line (§1.5), quadratics and curvature (§1.6), the exponential and the logarithm (§2.1–§2.2).
After it you will understand: what a derivative is and how to compute one, what a second derivative says about the shape of a curve, and what an integral accumulates. Chapter 4 lifts all of it to many variables, which is where machine learning actually lives.

Calculus is the mathematics of change. Everything in this thesis that *learns* does so by measuring how a number responds to a small nudge and then nudging in the direction that helps. This chapter builds that measurement for functions of one variable.

3.1 Limits, Informally

Situation — the everyday picture

A runner covers 100 metres in 10 seconds. Her average speed is 10 m/s, but she was not travelling at 10 m/s throughout — she started at zero. To find her speed at the exact instant $t = 4$ seconds, you might time her over the interval from 4 to 5 seconds. That is still an average. So you shrink the window: 4 to 4.1, then 4 to 4.01, then 4 to 4.001.

Problem — where it breaks or why it matters

You cannot shrink the window to nothing. Over an interval of exactly zero duration she covers exactly zero distance, and $0/0$ is not a number. Yet the sequence of averages over shrinking windows plainly settles down toward a particular value. We need language for "the value it settles toward" that does not require ever reaching a window of zero width.

Worked Arithmetic — concrete numbers

Take $f(x) = x^2$ and ask for the slope at $x = 3$.
Average slope from 3 to $3 + h$ is

$$\frac{f(3+h) - f(3)}{h} = \frac{(3+h)^2 - 9}{h} = \frac{9 + 6h + h^2 - 9}{h} = \frac{6h + h^2}{h} = 6 + h,$$

where the cancellation is legal for every $h \neq 0$.

h	Average slope $6+h$
1	7
0.1	6.1
0.01	6.01
0.001	6.001
-0.001	5.999
-0.1	5.9

Approaching from either side, the values close in on 6. At $h=0$ the expression $(6h+h^2)/h$ is $0/0$ and says nothing — but $6+h$ at $h=0$ is plainly 6. The limit is 6.

Formal Name — the mathematics

We write

$$\lim_{h\to 0} g(h) = L$$

to mean: the values $g(h)$ can be made as close to L as desired by taking h sufficiently close to (but not equal to) 0.

The limit describes the value g *approaches*; whether $g(0)$ exists, and what it equals if it does, is a separate question. In the example above $g(h) = (6h+h^2)/h$ is undefined at $h=0$ yet has limit 6 there.

Where the Analogy Breaks

Not every function settles. Consider $g(h) = 1/h$ near zero: from the right it grows without bound, from the left it falls without bound, and there is no value it approaches. Others approach different values from each side and so have no limit either. The limit is a claim that must be checked, not a formality — and the same care resurfaces in Chapter 9, where the sample kurtosis of a heavy-tailed series has no limit at all, drifting upward without settling as n grows.

Why It Matters Here — link to the thesis

Every derivative in this thesis is a limit of this kind. The autograd machinery in PyTorch does not compute limits numerically — it applies the rules of §3.3, which were themselves derived from limits once and for all.

3.2 The Derivative as Instantaneous Slope

Situation — the everyday picture

Your car's speedometer reads 60 km/h. It is not telling you a distance divided by a duration over any interval you could name — it is reporting how fast the odometer is changing *right now*. The dashboard shows one number, and it is the limit of the previous section made physical.

Problem — where it breaks or why it matters

For a straight line, one slope describes the whole thing (§1.5). For anything that bends, the slope differs at every point, so "the slope" is not a number but a new *function* of position.

Building that function is the central construction of calculus.

Worked Arithmetic — concrete numbers

The two-point picture first. For $f(x) = x^2$, from $x = 1$ to $x = 4$: heights 1 and 16, change 15, run 3, ratio 5. That is the average slope over the interval — the slope of the straight line joining the two points, called a *secant*.
Shrink the interval toward $x = 1$:

From	To	Rise	Run	Slope
1	4	15	3	5
1	2	3	1	3
1	1.1	0.21	0.1	2.1
1	1.01	0.0201	0.01	2.01

Settling on 2. Repeating the algebra of §3.1 at a general point:

$$\frac{(x+h)^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h \xrightarrow[h\to 0]{} 2x.$$

So the derivative of x^2 is $2x$: at $x = 1$ it is 2, at $x = 3$ it is 6 (matching §3.1), at $x = 0$ it is 0 — the flat bottom of the bowl.

Sanity check against §1.6. There the average slopes over unit steps were $1, 3, 5, 7$. The derivative at the *midpoints* $0.5, 1.5, 2.5, 3.5$ is $2x = 1, 3, 5, 7$. Exactly. ✓

Formal Name — the mathematics

The **derivative** of f at x is

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h},$$

when the limit exists. Equivalent notations: $f'(x)$, $\dfrac{df}{dx}$, $\dfrac{d}{dx}f(x)$.
$f'(x)$ is the slope of the **tangent** — the straight line touching the curve at x and matching its direction there. A function with a derivative everywhere on an interval is **differentiable** there.

Where the Analogy Breaks

Not every function has a derivative everywhere. The absolute value $|x|$ has slope -1 to the left of zero and $+1$ to the right, and no single tangent at the corner: the limit from each side disagrees. This is not an exotic pathology. The ReLU activation used throughout deep learning is exactly this shape, and the gradient penalty of Chapter 13 involves $\big|\,\|\nabla D\| - 1\big|$, which has the same kink. Practical systems patch the corner by picking a convention; the mathematics genuinely has no answer there.

Why It Matters Here — link to the thesis

Training any model in this thesis means computing $\partial(\text{loss})/\partial(\text{weight})$ for every weight and stepping against it. The gradient-clipping threshold of 1.0 applied in all three generators is a bound on the size of exactly these derivatives.

3.3 The Rules of Differentiation

Situation — the everyday picture

Nobody computes limits by hand to differentiate. Once the limit has been evaluated for a handful of basic functions, and once a few rules for combining them are established, differentiation becomes bookkeeping — which is precisely why a computer can do it automatically.

Problem — where it breaks or why it matters

The functions we actually differentiate are built by adding, multiplying, dividing and composing simpler ones. Without combination rules, each new combination would require returning to the limit definition from scratch.

Worked Arithmetic — concrete numbers

Power rule: $\frac{d}{dx}x^n = nx^{n-1}$.
Check on x^3 at $x = 2$. The rule gives $3x^2 = 12$. Numerically, with $h = 0.001$:

$$\frac{(2.001)^3 - 2^3}{0.001} = \frac{8.012006001 - 8}{0.001} = 12.006,$$

converging to 12 as h shrinks. ✓

Sum rule: $(f + g)' = f' + g'$. For $f(x) = x^2 + x^3$, $f'(x) = 2x + 3x^2$. At $x = 2$: $4 + 12 = 16$.

Product rule: $(fg)' = f'g + fg'$.
Take $f = x^2$, $g = x^3$, so $fg = x^5$ and the power rule predicts $5x^4$. Via the product rule: $2x \cdot x^3 + x^2 \cdot 3x^2 = 2x^4 + 3x^4 = 5x^4$. ✓
Note that $(fg)' \neq f'g'$: here $f'g' = 2x \cdot 3x^2 = 6x^3$, which is wrong.

Chain rule: $\big(f(g(x))\big)' = f'(g(x)) \cdot g'(x)$.
Take $f(u) = u^2$ and $g(x) = 3x + 1$, so $f(g(x)) = (3x + 1)^2$. The rule gives $2(3x + 1) \cdot 3 = 6(3x + 1) = 18x + 6$. Expanding first: $(3x + 1)^2 = 9x^2 + 6x + 1$, whose derivative is $18x + 6$. ✓

Exponential and logarithm:

$$\frac{d}{dx}e^x = e^x, \qquad \frac{d}{dx}\ln x = \frac{1}{x} \quad (x > 0).$$

The first is the property promised in §2.1: e^x is its own derivative, the unique function (up to scaling) that is its own rate of change.

Formal Name — the mathematics

For differentiable f, g and constant c:

$$\begin{aligned}
(cf)' &= cf' & (f \pm g)' &= f' \pm g' \\
(fg)' &= f'g + fg' & \left(\frac{f}{g}\right)' &= \frac{f'g - fg'}{g^2} \quad (g \neq 0) \\
(f \circ g)'(x) &= f'(g(x))\, g'(x) & \frac{d}{dx}x^n &= nx^{n-1}
\end{aligned}$$

The chain rule is the one that matters most: it is the mathematical content of backpropagation.

Where the Analogy Breaks

The rules make differentiation look purely mechanical, and symbolically it is. But mechanical is not the same as numerically harmless. Applying the chain rule through fifty composed layers multiplies fifty derivatives together, and if each is around 0.5 the product is about 10^{-15} — the vanishing-gradient problem. The algebra is exact; the arithmetic that implements it is not.

Why It Matters Here — link to the thesis

The chain rule applied to a deep composition is backpropagation, and it is why Chapter 4's multivariable version is the load-bearing piece of this whole book. The vanishing-gradient issue above is the direct reason this project replaced sigmoid activations with tanh in the TimeGAN latent path: $\tanh'(0) = 1$ against $\sigma'(0) = 0.25$, a fourfold stronger signal per layer.

3.4 Second Derivatives and Curvature

Situation — the everyday picture

You are in a car. The speedometer says how fast you are going; the push you feel in your back says how fast *that* is changing. Speed is the first derivative of position, acceleration the second. They answer different questions, and you can have one without the other: at constant 100 km/h the speed is large and the acceleration is zero.

Problem — where it breaks or why it matters

The first derivative alone cannot tell a valley from a peak. At the bottom of a bowl and at the top of a hill the slope is zero in both cases. Distinguishing them requires knowing which way the curve bends.

Worked Arithmetic — concrete numbers

Take $f(x) = x^2$, so $f'(x) = 2x$ and $f''(x) = 2$.
At $x = 0$: $f'(0) = 0$, so it is a turning point. $f''(0) = 2 > 0$, meaning the slope is increasing as we pass through — from negative, to zero, to positive. That is a valley.
Check directly: $f(-1) = 1$, $f(0) = 0$, $f(1) = 1$. Lower in the middle. ✓

Now $g(x) = -x^2$, so $g'(x) = -2x$ and $g''(x) = -2$.
At $x = 0$: $g'(0) = 0$ again, but $g''(0) = -2 < 0$. The slope is decreasing through the point — positive, zero, negative. A peak.
Check: $g(-1) = -1$, $g(0) = 0$, $g(1) = -1$. Higher in the middle. ✓

The ambiguous case. $h(x) = x^3$ has $h'(0) = 0$ and $h''(0) = 0$. It is neither peak nor valley: $h(-1) = -1$ and $h(1) = 1$, so the function passes straight through while momentarily flattening. When the second derivative vanishes, the test is silent.

Formal Name — the mathematics

The **second derivative** $f''(x)$ is the derivative of f'. It measures **concavity**:

- $f''(x) > 0$: convex here (curving upward, holds water);
- $f''(x) < 0$: concave here (curving downward, sheds water);
- $f''(x) = 0$: no information from this test.

Second-derivative test. If $f'(a) = 0$ and $f''(a) > 0$, then a is a local minimum; if $f'(a) = 0$ and $f''(a) < 0$, a local maximum.

Where the Analogy Breaks

"Local" is doing heavy lifting. The test inspects an infinitesimal neighbourhood and certifies nothing beyond it: a function may have a local minimum at a and still take far lower values somewhere else entirely. Every optimisation result in this thesis is local in exactly this sense — nothing in the training procedure certifies a global optimum, and nothing in the reported numbers should be read as claiming one.

Why It Matters Here — link to the thesis

Convexity is what makes an optimisation problem easy, and its absence is what makes GAN training hard. A convex loss has one basin and any downhill walk finds it. A GAN's objective is not convex, and moreover the two networks are optimising against each other, so the surface each is descending shifts as the other learns. This is the structural reason for the instability Chapter 13 documents, and for the mode collapse the post-generation guard in this project is built to detect.

3.5 Integrals as Accumulated Area

Situation — the everyday picture

A tap fills a bucket. The flow rate varies — fast at first, then slowing. You know the rate at every instant and want the total volume after ten minutes. If the rate were constant you would multiply rate by time. It is not, so you cannot.

Problem — where it breaks or why it matters

Differentiation goes from total to rate. The reverse trip — rate to total — is needed just as often, and it is not simply the same operation run backwards in any obvious way. It requires its own construction.

Worked Arithmetic — concrete numbers

Constant rate first. A tap running at 2 litres/minute for 3 minutes delivers $2 \times 3 = 6$ litres. On a graph of rate against time this is a rectangle of height 2 and width 3 — area 6. Total is area under the rate curve.

Varying rate. Let the rate be $f(t) = 2t$ litres/minute, from $t = 0$ to $t = 3$. Approximate with three rectangles of width 1, using the rate at the right edge of each:

$$f(1) \cdot 1 + f(2) \cdot 1 + f(3) \cdot 1 = 2 + 4 + 6 = 12.$$

Too big — the rate is rising, so the right edge overstates each strip. Using left edges:

$$f(0) + f(1) + f(2) = 0 + 2 + 4 = 6,$$

too small. The truth is between 6 and 12.

Refine to six strips of width 0.5, right edges:

$$0.5(f(0.5) + f(1) + \cdots + f(3)) = 0.5(1 + 2 + 3 + 4 + 5 + 6) = 0.5 \times 21 = 10.5.$$

Left edges give $0.5(0+1+2+3+4+5) = 7.5$. The bracket has narrowed from $[6, 12]$ to $[7.5, 10.5]$.

Exactly. The region under $f(t) = 2t$ from 0 to 3 is a triangle of base 3 and height 6, so its area is $\frac{1}{2} \times 3 \times 6 = 9$ litres — and 9 sits inside both brackets. ✓

Formal Name — the mathematics

The **definite integral** of f from a to b,

$$\int_a^b f(x)\,dx,$$

is the limit of such rectangle sums as the widths shrink to zero. It equals the signed area between the graph and the horizontal axis — signed because regions below the axis count negatively.
The elongated S is a stretched "sum", and dx records the vanishing width of each strip.

Where the Analogy Breaks

Area is the right picture in one dimension and becomes a misleading one quickly. In the probability chapters an integral over a two-dimensional region is a volume, and the integrals defining Wasserstein distance are over spaces of *joint distributions* — objects with no spatial interpretation at all. What survives from the picture is the accounting: an integral totals up infinitely many infinitesimal contributions. That much always holds.

Why It Matters Here — link to the thesis

A probability density is a rate — probability per unit of return — so probabilities are integrals of it. The statement that a 5σ day has probability 5.73×10^{-7} under a Gaussian is the value of an integral over the two tails, and that single number is what yields the once-per-6,922-years figure that motivates this entire thesis.

3.6 The Fundamental Theorem

Situation — the everyday picture

Your car has an odometer and a speedometer. The speedometer reading is the rate of change of the odometer. Equally, the odometer reading is the accumulation of the speedometer over the journey. Two instruments, one relationship, read in opposite directions.

Problem — where it breaks or why it matters

Computing integrals by shrinking rectangles is exhausting and, for most functions, impossible by hand. If accumulation really is the inverse of differentiation, then every derivative already known becomes an integral known for free — and §3.3 supplied a whole table of them.

Worked Arithmetic — concrete numbers

Redo the tap by reversing differentiation. We need a function whose derivative is $2t$. From the power rule, $\frac{d}{dt}t^2 = 2t$, so $F(t) = t^2$ works.

Then

$$\int_0^3 2t\,dt = F(3) - F(0) = 9 - 0 = 9,$$

matching the triangle exactly, with no rectangles at all. ✓

Second example. $\int_1^4 2x\,dx = F(4) - F(1) = 16 - 1 = 15.$
Geometric check: the region is a trapezium with parallel sides $f(1) = 2$ and $f(4) = 8$, width 3. Area $= \frac{1}{2}(2 + 8) \times 3 = 15$. ✓

Third. $\int_1^e \frac{1}{x}\,dx = \ln(e) - \ln(1) = 1 - 0 = 1$, since $\frac{d}{dx} \ln x = 1/x$. The area under $1/x$ from 1 to e is exactly 1 — which is one way of saying what the number e is.

Formal Name — the mathematics

Fundamental Theorem of Calculus. If $F' = f$ on $[a, b]$ and f is continuous, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Conversely, $\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$.
F is an **antiderivative** of f. Antiderivatives are not unique — any constant may be added, since constants differentiate to zero — but the constant cancels in $F(b) - F(a)$, so the definite integral is unaffected.

Where the Analogy Breaks

The theorem guarantees an antiderivative exists for any continuous f; it does not guarantee you can write one down. The Gaussian density $e^{-x^2/2}$ has no antiderivative expressible in elementary functions, which is exactly why the normal CDF Φ must be tabulated or computed numerically. The 5.73×10^{-7} quoted above is a numerical result, not a closed-form one.

Why It Matters Here — link to the thesis

The theorem links densities to probabilities throughout this thesis: the CDF is the integral of the density, and the density is the derivative of the CDF. Every quantile — and Value at Risk is precisely a quantile — is found by inverting that relationship.

Chapter 4

Calculus II — Many Variables

Before and After

Before this chapter you need: derivatives and the rules for computing them (§3.2–§3.3), second derivatives and concavity (§3.4), composition of functions (§1.7).
After it you will understand: what a partial derivative is, why the gradient points uphill, how the chain rule propagates through a deep composition — which is backpropagation — and what $\|\nabla D\|_2 = 1$ is asking for in the WGAN gradient penalty.

This is the load-bearing chapter of the book. Gradient descent, backpropagation and the gradient penalty are all direct applications of what follows. Everything here is the one-variable calculus of Chapter 3 applied one variable at a time, and then assembled.

A note on ordering: this chapter needs to collect several numbers into a single object and occasionally measure its length. It does so using only school geometry — Pythagoras — and defers the general theory of such objects to Chapter 5, which gives them their proper treatment.

4.1 Functions of Several Variables

Situation — the everyday picture

The temperature in a room depends on where you stand. Move toward the radiator and it rises; move toward the window and it falls. To state the temperature you must give two numbers — how far along, how far across — and you get back one number.

Problem — where it breaks or why it matters

A curve drawn on a page cannot represent this. Two inputs and one output need three dimensions, and the objects we actually optimise in this thesis have hundreds of thousands of inputs. Any intuition tied to pictures will fail; what survives is the arithmetic.

Worked Arithmetic — concrete numbers

Take $f(x, y) = x^2 + y^2$ — squared distance from the origin. Evaluate on a grid:

	$y=0$	$y=1$	$y=2$	$y=3$
$x=0$	0	1	4	9
$x=1$	1	2	5	10
$x=2$	4	5	8	13
$x=3$	9	10	13	18

Read across row $x = 2$: values $4, 5, 8, 13$, rising as y grows. Read down column $y = 1$: values $1, 2, 5, 10$, rising as x grows. The single lowest entry is 0 at the corner $(0, 0)$ — the bottom of a bowl, now in two input directions rather than one.

Notice the entries equal to 5: at $(1, 2)$ and at $(2, 1)$. Different points, same height. Joining all points of equal height gives a **contour**, exactly as on a map. Here the contours are circles about the origin.

Formal Name — the mathematics

A function of several variables assigns one output number to each combination of inputs, written $f : \mathbb{R}^n \to \mathbb{R}$. The notation $\mathbb{R}^n$ means ordered lists of n real numbers; $\mathbb{R}^2$ is pairs (x, y), $\mathbb{R}^3$ is triples, and so on.
A **level set** or **contour** is the collection of inputs sharing one output value: $\{(x, y) : f(x, y) = c\}$.

Where the Analogy Breaks

Two inputs can be pictured as a landscape; three already cannot, and the loss function of the TimeGAN generator in this project has 11,400 inputs — one per weight. Nothing visual survives. What does survive is that each input can be varied on its own while the rest are held fixed, and that is the entire content of the next section.

Why It Matters Here — link to the thesis

A neural network's loss is exactly such a function: inputs are the weights, output is a single number measuring how badly the network is doing. Training searches that space for a low point.

4.2 Partial Derivatives

Situation — the everyday picture

You are on a hillside at a crossroads. One road runs due east, another due north. Standing still, you can ask two separate questions: if I step east, do I go up or down, and how steeply? And the same for north. Two roads, two answers, one place.

Problem — where it breaks or why it matters

"The slope" of a surface is not a single number, because the answer depends on which way you face. But if we agree to ask only about the coordinate directions — one at a time, everything else frozen — each question reduces to the one-variable derivative already available.

Worked Arithmetic — concrete numbers

Take $f(x, y) = x^2 + y^2$ at the point $(3, 4)$, where $f = 9 + 16 = 25$.

Step east (x changes, y frozen at 4). The function becomes $g(x) = x^2 + 16$, a one-variable function. Its derivative is $2x$, which at $x = 3$ is 6.
Check numerically: $f(3.001, 4) = 9.006001 + 16 = 25.006001$, so $(25.006001 - 25)/0.001 = 6.001$, tending to 6. ✓

Step north (y changes, x frozen at 3). Now $h(y) = 9 + y^2$, derivative $2y$, which at $y = 4$ is 8.
Check: $f(3, 4.001) = 9 + 16.008001 = 25.008001$, giving $(25.008001 - 25)/0.001 = 8.001 \to 8$. ✓

The recipe. To differentiate with respect to one variable, treat every other variable as a constant. For $f(x, y) = x^2y^3$:

$$\frac{\partial f}{\partial x} = 2xy^3 \quad (y^3 \text{ is just a number}), \qquad \frac{\partial f}{\partial y} = 3x^2y^2 \quad (x^2 \text{ is just a number}).$$

At $(2, 1)$: $\partial f/\partial x = 2 \cdot 2 \cdot 1 = 4$ and $\partial f/\partial y = 3 \cdot 4 \cdot 1 = 12$.

Formal Name — the mathematics

The **partial derivative** of f with respect to x_i is

$$\frac{\partial f}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, \ldots, x_i + h, \ldots, x_n) - f(x_1, \ldots, x_n)}{h},$$

the ordinary derivative in the i-th coordinate with all others held fixed. The curved ∂ distinguishes it from the d of one-variable calculus and is a reminder that other variables exist and are being held still.

Where the Analogy Breaks

Partial derivatives only ever look along the coordinate axes. A ridge running diagonally can have zero slope due east and zero due north while dropping steeply to the southwest — both partials vanish and the point is still not a minimum. The coordinate directions are a convenient basis, not a complete description, and §4.3 is what repairs the gap.

Why It Matters Here — link to the thesis

$\partial(\text{loss})/\partial w$ for a single weight w is what an optimiser consumes. PyTorch's `backward()` computes exactly this for every weight in the network in one sweep.

4.3 The Gradient

Situation — the everyday picture

Back at the crossroads. You know the eastward slope is 6 and the northward slope is 8. You are not obliged to walk along a road — you may set off in any direction. Which one climbs fastest, and how steep is it?

Problem — where it breaks or why it matters

The partials give two numbers; the question asks for a direction and a steepness. Something must convert the pair into that, and the conversion had better be the one that actually identifies the steepest route, since gradient descent depends on going exactly opposite to it.

Worked Arithmetic — concrete numbers

Collect the partials into an ordered pair, written in brackets:

$$\nabla f(3,4) = [\,6,\ 8\,].$$

Claim: this pair points in the steepest uphill direction, and its length is the steepest slope.

Its length, by Pythagoras:

$$\sqrt{6^2+8^2} = \sqrt{36+64} = \sqrt{100} = 10.$$

Check that no direction beats 10. Moving a small distance h in a direction that is a fraction a east and b north (with $a^2+b^2=1$) changes f by approximately $h(6a+8b)$. Try several:

Direction (a,b)	$6a+8b$
Due east $(1,0)$	6
Due north $(0,1)$	8
Diagonal $(0.707, 0.707)$	$4.24+5.66=9.90$
$(0.6,0.8)$	$3.6+6.4=10.0$
$(0.5,0.866)$	$3.0+6.93=9.93$

The winner is $(0.6,0.8)$, giving exactly 10 — and $(0.6,0.8)$ is $[6,8]$ divided by its length 10. The gradient direction wins, and the best achievable rate equals the gradient's length. ✓

Downhill is the exact reverse, $[-6,-8]$, dropping at rate 10. That is the direction gradient descent steps in.

Formal Name — the mathematics

The **gradient** of $f:\mathbb{R}^n\to\mathbb{R}$ is the ordered list of its partial derivatives:

$$\nabla f = \left[\frac{\partial f}{\partial x_1},\ \frac{\partial f}{\partial x_2},\ \ldots,\ \frac{\partial f}{\partial x_n}\right].$$

Its **length** (Chapter 5 calls this the Euclidean or 2-norm) is

$$\|\nabla f\|_2 = \sqrt{\left(\frac{\partial f}{\partial x_1}\right)^2+\cdots+\left(\frac{\partial f}{\partial x_n}\right)^2}.$$

Two facts, both verified numerically above: ∇f points in the direction of steepest increase, and $\|\nabla f\|_2$ is the rate of increase in that direction. Where f has a local maximum or minimum in the interior of its domain, $\nabla f = \mathbf{0}$ — every partial vanishes at once.

Where the Analogy Breaks

Steepest *here* is not fastest *overall*. The gradient is purely local: it describes an infinitesimal neighbourhood and knows nothing of the terrain beyond. Repeatedly following it can zig-zag down a narrow valley, taking many small steps where one well-chosen direction would have done. It is greedy, and greedy is not optimal — which is why Chapter 10 (Optimisation)'s momentum and adaptive methods exist at all.

Why It Matters Here — link to the thesis

Two direct uses. First, gradient descent: every model here is trained by repeatedly stepping along $-\nabla(\text{loss})$. Second, and more specifically, the WGAN-GP gradient penalty is a statement about $\|\nabla D\|_2$ — it adds a cost whenever the length of the critic's gradient departs from 1. With that definition in hand, $\|\nabla D\|_2 = 1$ now reads concretely: whichever direction you perturb the input, the critic's output should change at a rate of exactly one unit per unit of perturbation. Chapter 13 explains why that is the right thing to ask for.

4.4 The Chain Rule and Backpropagation

Situation — the everyday picture

The factory line of §1.7, now with a quality inspector at the end who scores each finished item. A dial on station one is nudged. The painted part changes a little; the boxed part changes as a result; the score changes as a result of that. How much did the score move per unit of dial?

Problem — where it breaks or why it matters

The dial and the score are separated by every station in between. Measuring the effect by physically turning the dial and re-running the line is possible for one dial, but a network has hundreds of thousands, and re-running once per dial is hopelessly expensive. We need the whole answer from one pass.

Worked Arithmetic — concrete numbers

A chain of three. Let

$$u = g(x) = x + 3, \qquad v = f(u) = u^2, \qquad L = k(v) = 2v.$$

At $x = 1$: $u = 4$, $v = 16$, $L = 32$.
Local rates: $du/dx = 1$, $dv/du = 2u = 8$, $dL/dv = 2$.
Chain them by multiplying:

$$\frac{dL}{dx} = \frac{dL}{dv} \cdot \frac{dv}{du} \cdot \frac{du}{dx} = 2 \times 8 \times 1 = 16.$$

Verify by substitution. $L = 2(x+3)^2$, so $dL/dx = 4(x+3)$, which at $x = 1$ is 16. ✓
Verify numerically. At $x = 1.001$: $u = 4.001$, $v = 16.008001$, $L = 32.016002$. Then $(32.016002 - 32)/0.001 = 16.002 \to 16$. ✓

Two paths. Now let one quantity feed two others which recombine:

$$u = 2x, \qquad v = x^2, \qquad L = u + v.$$

At $x = 3$: $u = 6$, $v = 9$, $L = 15$.
$\partial L/\partial u = 1$ and $\partial L/\partial v = 1$; $du/dx = 2$ and $dv/dx = 2x = 6$. Influence arrives along both routes and the contributions *add*:

$$\frac{dL}{dx} = 1 \times 2 + 1 \times 6 = 8.$$

Check: $L = 2x + x^2$, so $dL/dx = 2 + 2x = 8$ at $x = 3$. ✓

The two operations are the whole algorithm: multiply along a path, add across paths.

Formal Name — the mathematics

If L depends on x through intermediate quantities $u_1, \ldots, u_m$, then

$$\frac{\partial L}{\partial x} = \sum_{j=1}^{m} \frac{\partial L}{\partial u_j} \frac{\partial u_j}{\partial x}.$$

Backpropagation evaluates this from the output backwards. A *forward pass* computes and stores every intermediate value; a *backward pass* starts with $\partial L/\partial L = 1$ and walks toward the inputs, multiplying by each local derivative and summing where paths meet.
Its efficiency is the point: one backward pass yields $\partial L/\partial w$ for *every* weight simultaneously, at roughly the cost of one forward pass — not one pass per weight.

Where the Analogy Breaks

The multiplication along a path is also the algorithm's weakness. Through n layers the result is a product of n local derivatives; if these are systematically below 1 the product collapses toward zero, and if above 1 it explodes. Neither is a flaw in the chain rule — the algebra is exact — but both are real obstacles in floating-point arithmetic, and both are why activation functions, initialisation schemes and gradient clipping receive so much attention.

Why It Matters Here — link to the thesis

This is how every model in this thesis is trained. It is also why the activation choice in TimeGAN mattered so much: with sigmoid in the latent path, the local derivative is at most 0.25, so three layers multiply to at most 0.0156 and the learning signal reaching the early layers is negligible. Switching to tanh, whose derivative reaches 1 at the origin, is what moved the autoencoder's R^2 from roughly zero to +0.995. The gradient-norm clip of 1.0 used everywhere in this project is a direct guard against the exploding case.

4.5 The Jacobian

Situation — the everyday picture

A machine takes two dials and drives three gauges. Turning either dial moves potentially all three gauges. To describe the machine's responsiveness you need one number for each dial–gauge pairing: two dials times three gauges is six numbers, naturally laid out in a grid.

Problem — where it breaks or why it matters

The gradient of §4.3 was defined for functions returning a single number. A neural network layer returns many numbers at once, so its responsiveness is not one list but a whole table of them — and the notation must keep track of which entry means what.

Worked Arithmetic — concrete numbers

Take $F(x, y) = (x + y,\ xy,\ x^2)$: two inputs, three outputs.
Write the outputs as $F_1 = x + y$, $F_2 = xy$, $F_3 = x^2$ and differentiate each with respect to each input:

$$\begin{aligned} \partial F_1/\partial x &= 1, & \partial F_1/\partial y &= 1, \\ \partial F_2/\partial x &= y, & \partial F_2/\partial y &= x, \\ \partial F_3/\partial x &= 2x, & \partial F_3/\partial y &= 0. \end{aligned}$$

Arrange as a grid, rows indexed by output and columns by input, evaluated at $(x, y) = (2, 3)$:

$$J = \begin{bmatrix} 1 & 1 \\ 3 & 2 \\ 4 & 0 \end{bmatrix}.$$

Reading it. Row 2, column 1 is 3: nudging x moves F_2 at rate 3. Check: $F_2 = xy$, and at $(2.001, 3)$ it is 6.003, up from 6, so the rate is 3. ✓
Row 3, column 2 is 0: y has no effect on $F_3 = x^2$ at all. ✓

Formal Name — the mathematics

For $F : \mathbb{R}^n \to \mathbb{R}^m$ with components $F_1, \ldots, F_m$, the **Jacobian** is the $m \times n$ grid

$$J = \begin{bmatrix} \partial F_1/\partial x_1 & \cdots & \partial F_1/\partial x_n \\ \vdots & \ddots & \vdots \\ \partial F_m/\partial x_1 & \cdots & \partial F_m/\partial x_n \end{bmatrix}, \qquad J_{ij} = \frac{\partial F_i}{\partial x_j}.$$

When $m = 1$ the Jacobian has a single row and is the gradient. The chain rule for composed multi-output functions is the product of their Jacobians, in the grid-multiplication sense defined in §5.2.

Where the Analogy Breaks

Jacobians are rarely built explicitly in practice. For a layer with 1,000 inputs and 1,000 outputs the grid holds a million entries, and a network has many such layers. Automatic differentiation instead computes the *product* of a Jacobian with a given list of numbers, which needs far less memory and is all backpropagation ever actually requires.

Why It Matters Here — link to the thesis

Every layer of every network here has a Jacobian, and backpropagation is the chained product of them. The critic's gradient in WGAN-GP is a one-row Jacobian — the critic outputs a single score — which is why ∇D is a gradient and its Euclidean length is the quantity the penalty constrains.

4.6 The Hessian

Situation — the everyday picture

A valley may be a narrow gorge or a broad shallow bowl. Both have a flat floor, so the gradient is zero in each. They demand completely different step sizes: the gorge punishes a long stride by throwing you up the opposite wall, while the bowl makes short strides a waste of time.

Problem — where it breaks or why it matters

The second-derivative test of §3.4 distinguished peaks from valleys with one number. With many inputs there is curvature in every direction and, worse, cross-terms describing how the slope in one direction changes as you move in another. One number cannot carry that.

Worked Arithmetic — concrete numbers

Take $f(x, y) = x^2 + 3y^2$. First partials:

$$\frac{\partial f}{\partial x} = 2x, \qquad \frac{\partial f}{\partial y} = 6y.$$

Both vanish only at $(0, 0)$, so that is the sole candidate turning point.
Now differentiate each partial with respect to each variable:

$$\frac{\partial^2 f}{\partial x^2} = 2, \quad \frac{\partial^2 f}{\partial y^2} = 6, \quad \frac{\partial^2 f}{\partial x \partial y} = 0, \quad \frac{\partial^2 f}{\partial y \partial x} = 0.$$

As a grid:

$$H = \begin{bmatrix} 2 & 0 \\ 0 & 6 \end{bmatrix}.$$

Reading it. Both diagonal entries are positive, so the surface curves upward in both coordinate directions: $(0, 0)$ is a minimum. The y direction curves three times as sharply as x, so the bowl is an elongated ellipse, steep across and shallow along.
Verify. $f(1, 0) = 1$ while $f(0, 1) = 3$. The same unit step costs three times as much in y. ✓

A saddle. For $g(x, y) = x^2 - y^2$ the gradient $[2x, -2y]$ also vanishes only at the origin, and

$$H = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}.$$

One positive, one negative: a minimum along x, a maximum along y. Neither peak nor valley but a saddle, and $g(1, 0) = 1$ against $g(0, 1) = -1$ confirms it. Such points are everywhere in high dimensions — with many directions available, it is far more likely that some curve up and others down than that all agree.

Formal Name — the mathematics

The **Hessian** of $f : \mathbb{R}^n \to \mathbb{R}$ is the $n \times n$ grid of second partial derivatives,

$$H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}.$$

For twice-continuously-differentiable f the order of differentiation does not matter, so $H_{ij} = H_{ji}$ and the grid is symmetric.

At a point where $\nabla f = \mathbf{0}$: if every direction curves upward the point is a local minimum; if every direction curves downward, a local maximum; if directions disagree, a saddle. Chapter 5 makes "every direction" precise via eigenvalues (§5.6).

Where the Analogy Breaks

The Hessian is the right diagnostic and the wrong computation. For a network with 100,000 weights it holds 10^{10} entries — tens of gigabytes — and must be recomputed at every step. No training procedure in this thesis forms one. Optimisers instead approximate curvature cheaply, which is exactly what Adam's per-parameter scaling amounts to (Chapter 10 (Optimisation)).

Why It Matters Here — link to the thesis

Curvature explains why a single global learning rate struggles: in a valley shaped like the $[2, 0; 0, 6]$ example, the step that suits one direction is wrong for the other. Adam — the optimiser used for every model in this project — addresses precisely this by maintaining a separate effective step size per parameter.

Chapter 5

Linear Algebra

Before and After

Before this chapter you need: functions and composition (§1.3, §1.7), linear functions and slope (§1.5), gradients and Jacobians (§4.3, §4.5).
After it you will understand: what a vector and a matrix are, why matrix multiplication is defined the way it is, what a norm measures, and what $\|\nabla D\|_2 = 1$ demands of the critic — the last piece needed before the gradient penalty of Chapter 13 can be read as a sentence rather than a formula.

Chapter 4 repeatedly collected numbers into lists and grids and promised a proper account later. This is that account. Neural networks are, at bottom, alternating stacks of matrix multiplications and simple non-linear functions, so the vocabulary here is the vocabulary in which every model in this thesis is written.

5.1 Scalars, Vectors and Matrices

Situation — the everyday picture

A shop tracks three products. Monday's sales are 4, 7 and 2 units. You could write three separate numbers, or you could write one row, $[4, 7, 2]$, and treat Monday as a single object. Add Tuesday, Wednesday and Thursday and you have a table — one row per day, one column per product.

Problem — where it breaks or why it matters

Handling each number separately means writing a separate line of algebra for each, and a network layer would need a thousand such lines. Treating the collection as one object lets a single symbol stand for the lot and a single operation act on all of it at once.

Worked Arithmetic — concrete numbers

Vectors. Monday $\mathbf{u} = [4, 7, 2]$, Tuesday $\mathbf{v} = [1, 0, 5]$.
Addition is entry by entry:

$$\mathbf{u} + \mathbf{v} = [4 + 1,\ 7 + 0,\ 2 + 5] = [5, 7, 7].$$

Total units sold across the two days, per product. The interpretation survives the operation, which is the sign that the operation is the right one.

Scaling multiplies every entry:

$$3\mathbf{u} = [12, 21, 6],$$

what three identical Mondays would produce.

The dot product. Suppose prices are $\mathbf{p} = [10, 2, 5]$ per unit. Monday's revenue is

$$\mathbf{u} \cdot \mathbf{p} = 4 \times 10 + 7 \times 2 + 2 \times 5 = 40 + 14 + 10 = 64.$$

Multiply matching entries, add the results. One number out of two lists.

Matrices. Stack the two days:

$$A = \begin{bmatrix} 4 & 7 & 2 \\ 1 & 0 & 5 \end{bmatrix},$$

a 2×3 matrix — 2 rows, 3 columns, always in that order. The entry $A_{23} = 5$ sits in row 2, column 3: Tuesday's sales of product 3.

Formal Name — the mathematics

A **scalar** is a single number. A **vector** is an ordered list of numbers; the number of entries is its **dimension**. A **matrix** is a rectangular array, of **size** $m \times n$ for m rows and n columns. For vectors of equal dimension, addition and scalar multiplication act entrywise. The **dot product** of $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i,$$

a scalar. Entry A_{ij} of a matrix is in row i, column j.

Where the Analogy Breaks

The spreadsheet picture makes a matrix look like passive storage. Most matrices in this thesis are not data at all — they are *actions*. A layer's weight matrix is a thing that transforms its input, and the next section defines multiplication precisely so that this reading is the correct one.

Why It Matters Here — link to the thesis

A window of 128 daily returns is a vector in $\mathbb{R}^{128}$; a batch of 64 such windows is a 64×128 matrix. The noise fed to the QuantGAN generator is a vector in $\mathbb{R}^{100}$. These are the literal objects the code moves around.

5.2 Matrix Multiplication as Composition

Situation — the everyday picture

Two currency conversions in sequence: pounds to euros, then euros to dollars. Each is a multiplication. Doing both is a single conversion from pounds to dollars, and its rate is the product of the two rates. The composite is itself a conversion of the same kind.

Problem — where it breaks or why it matters

Layers of a network apply in sequence, exactly like the conversions. If each layer is a matrix, the composition of two layers should again be a matrix — otherwise composing them would produce some new kind of object and the theory would not close. Matrix multiplication is *defined* so that it does close.

Worked Arithmetic — concrete numbers

Matrix times vector first. Let

$$A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}.$$

Each output entry is the dot product of a row of A with $\mathbf{x}$:

$$A\mathbf{x} = \begin{bmatrix} 2 \times 4 + 0 \times 5 \\ 1 \times 4 + 3 \times 5 \end{bmatrix} = \begin{bmatrix} 8 \\ 19 \end{bmatrix}.$$

Now compose. Let

$$B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}.$$

Apply A then B to $\mathbf{x}$:

$$B(A\mathbf{x}) = B \begin{bmatrix} 8 \\ 19 \end{bmatrix} = \begin{bmatrix} 1 \times 8 + 1 \times 19 \\ 0 \times 8 + 2 \times 19 \end{bmatrix} = \begin{bmatrix} 27 \\ 38 \end{bmatrix}.$$

The composite matrix. BA has entry (i, j) equal to the dot product of row i of B with column j of A:

$$BA = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 \cdot 2 + 1 \cdot 1 & 1 \cdot 0 + 1 \cdot 3 \\ 0 \cdot 2 + 2 \cdot 1 & 0 \cdot 0 + 2 \cdot 3 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 2 & 6 \end{bmatrix}.$$

Test it on $\mathbf{x}$:

$$(BA)\mathbf{x} = \begin{bmatrix} 3 \times 4 + 3 \times 5 \\ 2 \times 4 + 6 \times 5 \end{bmatrix} = \begin{bmatrix} 27 \\ 38 \end{bmatrix}. \quad \checkmark$$

Same answer. One matrix now does the work of two.

Order matters. Reversing gives

$$AB = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 1 & 7 \end{bmatrix} \neq BA.$$

Just as painting-then-boxing differed from boxing-then-painting in §1.7.

Formal Name — the mathematics

For A of size $m \times k$ and B of size $k \times n$, the product AB has size $m \times n$ with

$$(AB)_{ij} = \sum_{l=1}^{k} A_{il} B_{lj}.$$

The inner dimensions must match: an $m \times k$ can only multiply a $k \times n$.

Matrix multiplication is **associative**, $(AB)C = A(BC)$, and **distributive** over addition, but **not commutative**: $AB \neq BA$ in general.

Where the Analogy Breaks

Composition of matrices captures composition of *linear* maps only. A network built purely from matrices would collapse: stack fifty of them and the product is a single matrix, so fifty layers would have exactly the expressive power of one. The non-linear activation between layers is what prevents the collapse, and it is the entire reason deep networks are deeper than shallow ones.

Why It Matters Here — link to the thesis

A dense layer computes $W\mathbf{x} + \mathbf{b}$ — a matrix multiplication and a shift. The GRU cells in TimeGAN and the convolutions in QuantGAN and FinGAN are structured matrix multiplications. The parameter counts reported for this project's models (9,744 for the TimeGAN embedder, 11,400 for its generator) are counts of entries in these matrices.

5.3 Transpose, Identity and Inverse

Situation — the everyday picture

Undoing things. Multiplying by 5 is undone by dividing by 5, and doing both leaves the number untouched. Is there a matrix that undoes another, and a matrix that does nothing at all?

Problem — where it breaks or why it matters

Backpropagation runs a network backwards, and solving a linear system means undoing a transformation. Both need an account of reversal — including an account of when reversal is impossible, which turns out to be the more informative case.

Worked Arithmetic — concrete numbers

Transpose: flip rows and columns.

$$A = \begin{bmatrix} 4 & 7 & 2 \\ 1 & 0 & 5 \end{bmatrix}, \qquad A^\top = \begin{bmatrix} 4 & 1 \\ 7 & 0 \\ 2 & 5 \end{bmatrix}.$$

A 2×3 becomes 3×2; entry (i, j) moves to (j, i).

Identity: ones on the diagonal, zeros elsewhere.

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad I \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \cdot 4 + 0 \cdot 5 \\ 0 \cdot 4 + 1 \cdot 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}. \quad \checkmark$$

Inverse. For $A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$, the inverse is

$$A^{-1} = \begin{bmatrix} 0.5 & 0 \\ -1/6 & 1/3 \end{bmatrix}.$$

Check:

$$A^{-1}A = \begin{bmatrix} 0.5 \cdot 2 + 0 \cdot 1 & 0.5 \cdot 0 + 0 \cdot 3 \\ -\frac{1}{6} \cdot 2 + \frac{1}{3} \cdot 1 & -\frac{1}{6} \cdot 0 + \frac{1}{3} \cdot 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{3} + \frac{1}{3} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I. \quad \checkmark$$

When it fails. Take $C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, whose second row is twice the first. Then

$$C \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{and} \quad C \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Two different inputs, identical output. No rule could recover which one you started from, so no inverse exists. Information was destroyed — the same failure mode as multiplying an equation by zero in §1.2.

Formal Name — the mathematics

The **transpose** $A^\top$ satisfies $(A^\top)_{ij} = A_{ji}$, and $(AB)^\top = B^\top A^\top$ — the order reverses.
The **identity** I_n has $I_{ii} = 1$ and $I_{ij} = 0$ for $i \neq j$; it satisfies $AI = IA = A$.
A square A is **invertible** if some A^{-1} has $AA^{-1} = A^{-1}A = I$. Such an A^{-1} is unique when it exists. A matrix with no inverse is **singular**, and A is singular exactly when some non-zero vector satisfies $A\mathbf{x} = \mathbf{0}$.

Where the Analogy Breaks

Invertibility is a yes/no question in theory and a matter of degree in practice. A matrix can be technically invertible while being so close to singular that computing A^{-1} amplifies rounding error catastrophically. Numerical work therefore speaks of *conditioning* — how nearly singular a matrix is — rather than treating invertibility as a clean binary.

Why It Matters Here — link to the thesis

The transpose appears throughout backpropagation: if a layer's forward pass multiplies by W, the corresponding backward pass multiplies by $W^\top$. Singularity is also the right way to understand mode collapse — a generator that maps many different noise inputs to the same output has destroyed information in exactly the way C above does, and no amount of downstream processing can recover the diversity that was lost.

5.4 The Determinant

Situation — the everyday picture

Take the unit square — corners at $(0,0)$, $(1,0)$, $(0,1)$, $(1,1)$, area 1 — and apply a matrix to each corner. You get a parallelogram. Its area tells you the factor by which this transformation inflates or shrinks regions.

Problem — where it breaks or why it matters

We need a single number answering "does this transformation destroy information?". Squashing a two-dimensional region flat onto a line reduces its area to zero, and that is exactly the situation in which the map cannot be undone.

Worked Arithmetic — concrete numbers

A stretch. $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ sends $(1,0) \mapsto (2,0)$ and $(0,1) \mapsto (0,3)$. The unit square becomes a 2×3 rectangle of area 6.
Formula for 2×2: $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$. Here $2 \times 3 - 0 \times 0 = 6$. ✓

A shear. $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ sends $(1,0) \mapsto (1,0)$ and $(0,1) \mapsto (1,1)$. The square becomes a parallelogram with base 1 and height 1 — area still 1. And $\det B = 1 \times 1 - 1 \times 0 = 1$. ✓A shear slides without changing area.

A collapse. $C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ from §5.3 sends $(1,0) \mapsto (1,2)$ and $(0,1) \mapsto (2,4)$. Both images lie on the same line through the origin — $(2,4)$ is exactly twice $(1,2)$. The square is flattened onto a line, area zero. And $\det C = 1 \times 4 - 2 \times 2 = 0$. ✓
The determinant being zero and the inverse failing to exist are the same fact.

Formal Name — the mathematics

The **determinant** $\det A$ of a square matrix is the signed factor by which it scales volume. For 2×2,

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

Key properties:

- $\det A = 0$ if and only if A is singular;
- $\det(AB) = \det A \cdot \det B$;
- a negative determinant means orientation is flipped, like a mirror.

Where the Analogy Breaks

The volume picture is exact but the computation does not scale. Evaluating a determinant by the schoolbook cofactor expansion costs on the order of $n!$ operations — for $n = 20$ that is 10^{18}, beyond reach. Practical algorithms use factorisations costing about n^3 instead. Never mistake a clean definition for a usable algorithm.

Why It Matters Here — link to the thesis

Determinants govern how probability densities transform under an invertible map, via the Jacobian determinant — the mechanism underpinning normalising flows, which Chapter 12 (Generative Models) contrasts with GANs. GANs deliberately avoid this machinery, which is what frees them from any invertibility requirement and also why they provide no likelihood.

5.5 Norms

Situation — the everyday picture

You are at a street corner and a friend is three blocks east and four blocks north. How far away is the friend? If you can fly, five blocks — straight line. If you must follow streets, seven — three plus four. Both are honest answers to "how far"; they differ because they measure different things.

Problem — where it breaks or why it matters

"How big is this vector?" has no single answer, and the gradient penalty in WGAN-GP asks precisely that question about ∇D. Before that constraint can be understood, the notion of size must be pinned down — including which one is meant.

Worked Arithmetic — concrete numbers

Take $\mathbf{v} = [3, 4]$.

Euclidean (2-norm) — straight line:

$$\|\mathbf{v}\|_2 = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

Manhattan (1-norm) — along the streets:

$$\|\mathbf{v}\|_1 = |3| + |4| = 7.$$

Maximum (∞-norm) — the largest single component:

$$\|\mathbf{v}\|_\infty = \max(|3|, |4|) = 4.$$

Three sizes for one vector: 5, 7, 4. Always state which.

A three-entry example. $\mathbf{w} = [1, -2, 2]$:

$$\|\mathbf{w}\|_2 = \sqrt{1 + 4 + 4} = \sqrt{9} = 3, \qquad \|\mathbf{w}\|_1 = 1 + 2 + 2 = 5, \qquad \|\mathbf{w}\|_\infty = 2.$$

Scaling. Doubling a vector doubles every norm: $\|2\mathbf{w}\|_2 = \sqrt{4 + 16 + 16} = \sqrt{36} = 6 = 2 \times 3$. ✓

Unit length. Dividing by the 2-norm gives length exactly 1: $\mathbf{v}/5 = [0.6, 0.8]$, and $\sqrt{0.36 + 0.64} = 1$. ✓This is the vector that won the steepest-direction contest in §4.3 — now recognisable as the gradient rescaled to unit length.

Formal Name — the mathematics

A **norm** $\|\cdot\|$ assigns a non-negative size to each vector, subject to three requirements:

1. $\|\mathbf{v}\| = 0$ only when $\mathbf{v} = \mathbf{0}$;
2. $\|c\mathbf{v}\| = |c|\,\|\mathbf{v}\|$ for any scalar c;
3. $\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$ (the triangle inequality: no detour is shorter than going direct).

The p-**norm** is

$$\|\mathbf{v}\|_p = \left(\sum_i |v_i|^p\right)^{1/p},$$

giving the Manhattan norm at $p = 1$, the Euclidean at $p = 2$, and the maximum norm in the limit $p \to \infty$. Unsubscripted $\|\cdot\|$ conventionally means the Euclidean norm.

Where the Analogy Breaks

The choice of norm is not cosmetic. In high dimensions the norms diverge sharply: for a vector of 10,000 entries each equal to 1, the 2-norm is 100, the 1-norm is 10,000, and the ∞-norm is 1. A constraint that is mild in one may be severe in another, so "the gradient is bounded" means nothing until the norm is named.

Why It Matters Here — link to the thesis

This is the last piece the gradient penalty needed. The WGAN-GP objective adds

$$\lambda\,\mathbb{E}\Big[\big(\|\nabla D(\hat{x})\|_2 - 1\big)^2\Big]$$

to the critic's loss, with $\lambda = 10$ throughout this project. Read plainly: compute the critic's gradient at a point, measure its Euclidean length, and penalise the square of its deviation from 1. Chapter 13 explains why 1 is the target — it enforces the Lipschitz condition that makes the critic's output a valid Wasserstein estimate. The gradient-norm clipping applied to every model here, `max_norm=1.0`, is a separate use of the same 2-norm: rescale the whole gradient whenever its Euclidean length exceeds 1.

5.6 Eigenvalues and Eigenvectors

Situation — the everyday picture

Stretch a rubber sheet: twice as wide, three times as tall. Most drawn arrows come out pointing somewhere new. But an arrow lying exactly horizontal stays horizontal, merely twice as long; a vertical one stays vertical, three times as long. Those two directions are special to this particular stretch.

Problem — where it breaks or why it matters

A general matrix mixes every direction into every other, which makes repeated application hard to reason about. If we can find the directions the matrix merely scales, its behaviour becomes transparent: along those directions it is nothing but multiplication by a number.

Worked Arithmetic — concrete numbers

Take $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$.

$$A\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad A\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix} = 3\begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Directions unchanged, lengths scaled by 2 and 3. Eigenvalues 2 and 3.

A generic direction is not preserved:

$$A\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix},$$

which is not a multiple of $[1, 1]$ — the arrow has rotated.

A less obvious case. $B = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$.

Try $[1, 0]$: $B[1, 0]^\top = [3, 0]^\top = 3[1, 0]^\top$. Eigenvalue 3. ✓
Try $[1, -1]$: $B[1, -1]^\top = [3 - 1,\ -2]^\top = [2, -2]^\top = 2[1, -1]^\top$. Eigenvalue 2. ✓

Note in both examples the eigenvalues are the diagonal entries — true for triangular matrices, not in general.

Formal Name — the mathematics

A non-zero vector $\mathbf{v}$ is an **eigenvector** of a square matrix A with **eigenvalue** λ if

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The direction is preserved; only the length changes, by the factor λ (reversing if $\lambda < 0$). Eigenvalues solve $\det(A - \lambda I) = 0$. A real matrix need not have real eigenvalues — a rotation has none, since it preserves no direction — but a *symmetric* real matrix always does.

Where the Analogy Breaks

Eigenvectors give a complete picture only when there are enough of them to span the space, which is not guaranteed. Some matrices are *defective* and cannot be diagonalised at all. The convenient picture — "every matrix is just scaling along some special axes" — is a property of a well-behaved subclass, not a universal truth.

Why It Matters Here — link to the thesis

The Hessian of §4.6 is symmetric, so its eigenvalues are real, and they make "curves upward in every direction" precise: all eigenvalues positive means a minimum, all negative a maximum, mixed signs a saddle. Their ratio is the condition number, which measures how elongated the valley is and therefore how badly a single global learning rate will perform — the quantitative version of the gorge-versus-bowl problem that motivates Adam.

5.7 Vector Spaces and Linear Transformations

Situation — the everyday picture

Recipes. Combine half a batch of one and two batches of another and you get a sensible recipe. Combine sound waves and you get a sound; combine forces and you get a force. In each case adding two objects and scaling one produce objects of the same kind. The collection is closed under those two operations.

Problem — where it breaks or why it matters

The same theorems keep reappearing for lists of numbers, for functions, for random variables. Rather than reproving each time, we identify what these settings share — and then prove things once, for anything that shares it.

Worked Arithmetic — concrete numbers

Is the set of vectors $[x, y]$ with $y = 2x$ a vector space?
Take $[1, 2]$ and $[3, 6]$, both on the line. Their sum is $[4, 8]$, and $8 = 2 \times 4$. ✓Still on the line.
Scale $[1, 2]$ by -5: $[-5, -10]$, and $-10 = 2 \times (-5)$. ✓
Zero: $[0, 0]$ satisfies $0 = 2 \times 0$. ✓
Closed under both operations and containing zero — yes.

Is the set with $y = 2x + 1$?
Take $[0, 1]$ and $[1, 3]$, both on the line. Their sum is $[1, 4]$, but $2 \times 1 + 1 = 3 \neq 4$. $\times$ Not closed. Nor does it contain $[0, 0]$, since $2 \times 0 + 1 = 1 \neq 0$.
The offset $+1$ is fatal, and this is exactly why $f(x) = mx + c$ with $c \neq 0$ is called *affine* rather than linear (§1.5).

Is a matrix map linear? For $A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$, $\mathbf{u} = [1, 0]$, $\mathbf{v} = [0, 1]$:

$$A(\mathbf{u} + \mathbf{v}) = A[1, 1]^\top = [2, 4]^\top,$$

$$A\mathbf{u} + A\mathbf{v} = [2, 1]^\top + [0, 3]^\top = [2, 4]^\top. \quad \checkmark$$

Formal Name — the mathematics

A **vector space** is a set closed under addition and scalar multiplication, containing a zero element, with each element having an additive inverse, and satisfying the usual associative, commutative and distributive laws.
A map T between vector spaces is a **linear transformation** if

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \quad \text{and} \quad T(c\mathbf{v}) = cT(\mathbf{v}).$$

Every linear transformation between finite-dimensional spaces is a matrix multiplication, once bases are fixed — which is why matrices and linear maps are so often treated as the same thing.
A **basis** is a minimal set of vectors from which every element can be built by addition and scaling; the number of them is the **dimension**.

Where the Analogy Breaks

Almost nothing interesting in this thesis is linear. Activation functions are deliberately non-linear, GAN losses are non-convex, and the map from noise to a synthetic return series is wildly non-linear by design. Linear algebra is not a description of these systems — it is the local language in which each small piece is written, and the thing calculus reduces them to one step at a time.

Why It Matters Here — link to the thesis

Every layer is a linear map followed by a non-linearity, and this alternation is what gives a deep network its expressive power: without the non-linearity the whole stack would collapse to a single matrix (§5.2). The set of returns a generator can produce is not a vector space — it is a curved subset of $\mathbb{R}^{128}$ — and Chapter 12 (Generative Models) takes up what that means for the manifold a generative model is trying to learn.

Chapter 6

Probability

Before and After

Before this chapter you need: fractions and percentages (§1.1), functions and their range (§1.3–§1.4), the exponential and logarithm (Chapter 2), integrals as accumulated area (§3.5), vectors and the dot product (§5.1).
After it you will understand: what a random variable is, how densities and probabilities relate, what expectation and variance actually average, why independence is the assumption market returns violate, and — the point the whole thesis turns on — why the Central Limit Theorem does not rescue you when the fourth moment does not exist.

Probability is the mathematics of uncertainty. This chapter builds it from counting outcomes up to the two limit theorems, and then shows precisely where those theorems stop applying to financial returns.

6.1 Sets and Events

Situation — the everyday picture

Roll one die. Six things can happen. You might care about "a six", or about "an even number", or about "more than four". Each of these is a collection of outcomes: $\{6\}$, $\{2,4,6\}$, $\{5,6\}$. The question you ask picks out a collection.

Problem — where it breaks or why it matters

Before anything can be assigned a probability, we need to say clearly what the things being assigned probabilities *are*. "The chance it rains" is ambiguous until "it rains" is pinned to a definite collection of outcomes, and ambiguity here produces paradoxes later.

Worked Arithmetic — concrete numbers

Rolling one die, the complete list of outcomes is

$$\Omega = \{1,2,3,4,5,6\}.$$

Let $A =$ "even" $= \{2,4,6\}$ and $B =$ "more than four" $= \{5,6\}$.

Both happen (intersection): $A \cap B = \{6\}$ — one outcome.
Either happens (union): $A \cup B = \{2,4,5,6\}$ — four outcomes.

A **does not happen** (complement): $A^c = \{1, 3, 5\}$ — three outcomes.

Counting check. $|A| = 3$, $|B| = 2$, $|A \cap B| = 1$, and

$$|A \cup B| = |A| + |B| - |A \cap B| = 3 + 2 - 1 = 4. \quad \checkmark$$

Subtracting the intersection is necessary because the outcome 6 lies in both sets and would otherwise be counted twice.

Formal Name — the mathematics

The **sample space** Ω is the set of all possible outcomes. An **event** is a subset of Ω. For events $A, B \subseteq \Omega$:

- $A \cap B$ (intersection): both occur;
- $A \cup B$ (union): at least one occurs;
- A^c (complement): A does not occur;
- $A \cap B = \emptyset$: the events are **disjoint** or **mutually exclusive** — they cannot both occur.

Where the Analogy Breaks

A die has six outcomes and every subset is a sensible event. For continuous quantities this breaks down: if Ω is the real line, some subsets are so pathological that no consistent probability can be assigned to them. Measure theory exists to exclude those, restricting attention to a well-behaved family of events. Nothing in this thesis meets such a set, but the restriction is why the formal definition below speaks of a chosen family rather than all subsets.

Why It Matters Here — link to the thesis

"A big move today" and "a big move yesterday" are the two events whose relationship measures volatility clustering. Making them precise — a return exceeding two standard deviations in absolute value — is what turns an intuition into something a generator can be scored against.

6.2 The Probability Axioms

Situation — the everyday picture

You are asked to bet on a horse race. Someone quotes you odds implying a 60% chance for one horse and a 70% chance for another, and there are five horses. You do not need to know anything about horses to know these numbers are broken: they already sum to more than certainty.

Problem — where it breaks or why it matters

Assigning numbers to events is only useful if the assignment is coherent. Three rules are enough to rule out every incoherent assignment, and everything else in probability follows from them.

Worked Arithmetic — concrete numbers

Fair die, each outcome probability 1/6.

Non-negative: every outcome has probability $1/6 > 0$. ✓
Total is one: $6 \times \frac{1}{6} = 1$. ✓
Disjoint events add: $A = \{2, 4, 6\}$ and $C = \{1, 3\}$ share nothing, so

$$\mathbb{P}(A \cup C) = \mathbb{P}(\{1, 2, 3, 4, 6\}) = \tfrac{5}{6},$$

and $\mathbb{P}(A) + \mathbb{P}(C) = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$. ✓
Overlapping events do not simply add. With $B = \{5, 6\}$:

$$\mathbb{P}(A) + \mathbb{P}(B) = \tfrac{3}{6} + \tfrac{2}{6} = \tfrac{5}{6},$$

but $A \cup B = \{2, 4, 5, 6\}$ has probability $\frac{4}{6}$. The excess $\frac{1}{6}$ is exactly $\mathbb{P}(A \cap B) = \mathbb{P}(\{6\})$, so

$$\mathbb{P}(A \cup B) = \tfrac{3}{6} + \tfrac{2}{6} - \tfrac{1}{6} = \tfrac{4}{6}. \quad \checkmark$$

Complement. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A) = 1 - \frac{3}{6} = \frac{3}{6}$, and $A^c = \{1, 3, 5\}$ indeed has three outcomes. ✓

Formal Name — the mathematics

A **probability measure** $\mathbb{P}$ assigns a number to each event subject to:

1. $\mathbb{P}(A) \geq 0$ for every event A;
2. $\mathbb{P}(\Omega) = 1$;
3. if $A_1, A_2, \ldots$ are pairwise disjoint, then $\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i)$.

These are the **Kolmogorov axioms**. Immediate consequences: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$, $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(A) \leq 1$, and the **inclusion–exclusion** rule

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Where the Analogy Breaks

The axioms say what a coherent assignment looks like. They say nothing about which assignment is *correct* for any real situation — that is a modelling question, answered by data or judgement, not by mathematics. A Gaussian model of returns satisfies every axiom perfectly and is still badly wrong about markets, as §6.12 shows.

Why It Matters Here — link to the thesis

The axioms are what make the shuffled-real control a valid baseline. Shuffling permutes outcomes without changing which values occur, so every probability depending only on the collection of values is untouched — which is precisely why the control scores a Wasserstein distance of exactly 0.000e+00 against real data while failing every ordering-sensitive test.

6.3 Conditional Probability

Situation — the everyday picture

A city has a 60% chance of rain on any given day. But if it rained yesterday, the chance today is 80%. Knowing yesterday's outcome changes the probability you assign to today's.

Problem — where it breaks or why it matters

Unconditional probabilities describe the long run and are silent about the present. A risk model that knows only that big moves happen 5% of the time, with no way to use the fact that yesterday was violent, is discarding most of what the data contains.

Worked Arithmetic — concrete numbers

Let A = "big market move today", B = "big market move yesterday".
MOEX, 20-year data: $\mathbb{P}(A) = 3.8\%$ while $\mathbb{P}(A \mid B) = 22.9\%$.

Ratio: $22.9/3.8 = 6.0$. Knowing yesterday was violent multiplies today's probability by roughly six.

Where the formula comes from. Conditioning on B restricts attention to the days when B happened, so B becomes the new sample space. Of those days, we ask what fraction also had A:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

On the die. $A = \{2, 4, 6\}$, $B = \{5, 6\}$:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(\{6\})}{\mathbb{P}(\{5, 6\})} = \frac{1/6}{2/6} = \tfrac{1}{2}.$$

Check by counting: told the roll exceeded four, it is 5 or 6, one of which is even. One in two. ✓

Note the asymmetry: $\mathbb{P}(B \mid A) = \frac{1/6}{3/6} = \frac{1}{3} \neq \frac{1}{2}$.

Formal Name — the mathematics

For $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Rearranging gives the **multiplication rule** $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\,\mathbb{P}(B)$.
Events A and B are **independent** when $\mathbb{P}(A \mid B) = \mathbb{P}(A)$, equivalently $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

Where the Analogy Breaks

Conditioning is not causation. MOEX's clustering after February 2022 is driven by a stream of news, not by the previous day's move causing the next. And the relationship is about *magnitude*, not direction: knowing yesterday was violent says a great deal about how large today's move will be and essentially nothing about its sign. That asymmetry is the whole content of volatility clustering.

Why It Matters Here — link to the thesis

The ratio $\mathbb{P}(A \mid B)/\mathbb{P}(A) = 6.0$ is a direct measurement of volatility clustering, and reproducing it is what separates a generator that has learned market dynamics from one that has only learned the histogram. The shuffled control destroys exactly this structure: after permutation $\mathbb{P}(A \mid B) \approx \mathbb{P}(A)$, the ratio collapses to 1, and the control fails every temporal metric while acing every fidelity one.

6.4 Bayes' Theorem

Situation — the everyday picture

A test for a rare disease is 99% accurate. You test positive. The intuitive reading — "99% chance I have it" — is wrong, and not slightly. If the disease affects one person in a thousand, the true figure is closer to 9%.

Problem — where it breaks or why it matters

People reliably confuse $\mathbb{P}(\text{positive} \mid \text{ill})$ with $\mathbb{P}(\text{ill} \mid \text{positive})$. The previous section already showed these differ; Bayes' theorem says exactly how, and the gap is largest precisely when the underlying event is rare — which is the regime every tail-risk question lives in.

Worked Arithmetic — concrete numbers

Take 100,000 people, one in a thousand ill, test 99% accurate in both directions.

	Ill	Healthy	Total
Test positive	99	999	1,098
Test negative	1	98,901	98,902
Total	100	99,900	100,000

Ill: 100 people, of whom 99% test positive → 99.
Healthy: 99,900 people, of whom 1% test positive → 999 false alarms.

Given a positive test:

$$\mathbb{P}(\text{ill} \mid +) = \frac{99}{99 + 999} = \frac{99}{1098} = 0.0902 = 9.0\%.$$

The false alarms outnumber the true positives ten to one, because there are a thousand times more healthy people to generate them. ✓

Via the formula:

$$\mathbb{P}(\text{ill} \mid +) = \frac{\mathbb{P}(+ \mid \text{ill})\mathbb{P}(\text{ill})}{\mathbb{P}(+)} = \frac{0.99 \times 0.001}{0.01098} = 0.0902. \quad ✓$$

Formal Name — the mathematics

Bayes' theorem. For events with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\,\mathbb{P}(A)}{\mathbb{P}(B)}.$$

The denominator expands by the **law of total probability**:

$$\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A) + \mathbb{P}(B \mid A^c)\mathbb{P}(A^c).$$

$\mathbb{P}(A)$ is the **prior**, $\mathbb{P}(A \mid B)$ the **posterior**, and $\mathbb{P}(B \mid A)$ the **likelihood**.

Where the Analogy Breaks

Bayes' theorem is an identity, not a philosophy. It tells you how to update given a prior; it does not tell you what prior to hold, and where the prior is contested the theorem settles nothing. The disease example is clean only because the base rate of one in a thousand was handed over as fact.

Why It Matters Here — link to the thesis

Base rates dominate rare-event inference, and every tail question in this thesis is a rare-event question. A test that flags a bad generator with 99% accuracy still produces mostly false alarms if bad generators are rare — which is the same arithmetic that makes multiple testing across 19 metrics (§7.9) so treacherous.

6.5 Random Variables

Situation — the everyday picture

Roll two dice and add them. The outcome of the experiment is a pair — say $(3, 4)$ — but what you care about is the single number 7. You have attached a number to each outcome.

Problem — where it breaks or why it matters

Events are collections of outcomes, and outcomes are often not numbers: heads or tails, a pair of dice, a market state. To do arithmetic — averages, spreads, distances — we need numbers, and we need a disciplined way of attaching them.

Worked Arithmetic — concrete numbers

Two dice, 36 equally likely outcomes. Let X be the sum.

X	2	3	4	5	6	7	8	9	10	11	12
Ways	1	2	3	4	5	6	5	4	3	2	1

The counts total $1 + 2 + 3 + 4 + 5 + 6 + 5 + 4 + 3 + 2 + 1 = 36$. ✓
$X = 7$ arises six ways — $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$ — so $\mathbb{P}(X = 7) = 6/36 = 1/6$, while $\mathbb{P}(X = 2) = 1/36$.

X is a *function*: it takes an outcome and returns a number, exactly the machine of §1.3. What makes it *random* is that its input is the result of a chance experiment, not the function itself.

Formal Name — the mathematics

A **random variable** is a function $X : \Omega \to \mathbb{R}$ assigning a real number to each outcome. It is **discrete** if its range is finite or countable, **continuous** if it takes values across an interval.

We write $\mathbb{P}(X = x)$ for the probability of the event $\{\omega \in \Omega : X(\omega) = x\}$, and $\mathbb{P}(X \leq x)$ similarly. Capital letters denote the random variable, lower case a particular value it might take.

Where the Analogy Breaks

"Random variable" is a poor name twice over: it is not a variable, and it is not random. It is an ordinary, fully determined function. All the randomness lives in which outcome the experiment produces — the function merely translates that outcome into a number.

Why It Matters Here — link to the thesis

The daily log return r_t is a random variable. So is every statistic computed from it: the Hill estimator, the sample kurtosis, the discriminative AUC. Treating these as random variables rather than fixed numbers is what makes it meaningful to ask about their sampling variability — the subject of Chapter 7.

6.6 PMF, PDF and CDF

Situation — the everyday picture

Two ways to describe rainfall. You could say how likely each amount is — "2 mm is common, 40 mm is rare". Or you could say how likely it is to be *at most* a given amount — "90% of days see 10 mm or less". Both describe the same weather.

Problem — where it breaks or why it matters

For discrete quantities, listing the probability of each value works. For continuous ones it collapses: the probability of a return being exactly $0.0123456\ldots$ is zero, and that is true of every single value, so the list carries no information. Continuous quantities need a different device.

Worked Arithmetic — concrete numbers

Discrete: the PMF. For the two-dice sum, $\mathbb{P}(X = 2) = 1/36$, $\mathbb{P}(X = 7) = 6/36$, and the values sum to 1.

Discrete: the CDF. Accumulate:

$$F(4) = \mathbb{P}(X \leq 4) = \tfrac{1}{36} + \tfrac{2}{36} + \tfrac{3}{36} = \tfrac{6}{36} = \tfrac{1}{6}.$$

And $F(12) = 1$, since every outcome is at most 12.

Continuous: why point probabilities vanish. Pick a real number uniformly from $[0, 1]$. The chance it lands in $[0.3, 0.4]$ is 0.1; in $[0.30, 0.31]$ is 0.01; in an interval of width w is w. Shrink w to zero and the probability goes to zero. Any single point has probability exactly zero, yet some point certainly occurs.

Continuous: the density. What survives is probability *per unit width*. For the uniform on $[0, 1]$ the density is $f(x) = 1$ throughout, and

$$\mathbb{P}(0.3 \leq X \leq 0.4) = \int_{0.3}^{0.4} 1\, dx = 0.4 - 0.3 = 0.1. \quad \checkmark$$

The density is a rate; the probability is its integral (§3.5). This is why a density may exceed 1 — a uniform on $[0, 0.5]$ has $f(x) = 2$ — without anything being wrong: it is a rate, not a probability.

Formal Name — the mathematics

For a discrete X, the **probability mass function** is $p(x) = \mathbb{P}(X = x)$, with $p(x) \geq 0$ and $\sum_x p(x) = 1$.
For a continuous X, the **probability density function** f satisfies

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)\,dx, \qquad f(x) \geq 0, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1.$$

In both cases the **cumulative distribution function** is

$$F(x) = \mathbb{P}(X \leq x),$$

non-decreasing, with $F(-\infty) = 0$ and $F(+\infty) = 1$. For continuous X the two are linked by the Fundamental Theorem of Calculus (§3.6): $F' = f$, and F is the integral of f.

Where the Analogy Breaks

A density is not a probability and cannot be compared across different quantities without care: change the units of x and f changes too, since it is measured per unit of x. The CDF has no such problem — it is a probability, and dimensionless. This is one reason distances between distributions in this thesis are built from CDFs and quantiles rather than densities.

Why It Matters Here — link to the thesis

The Kolmogorov–Smirnov statistic is the largest vertical gap between two CDFs. Value at Risk is an inverse CDF — the return below which only 5% of days fall. And the histograms in the stylised-facts figure are estimates of f; the log-scaled panel (§2.5) is the one that makes the tail of f legible.

6.7 Expectation

Situation — the everyday picture

A fairground game costs \$2. You win \$10 with probability 1/8 and nothing otherwise. Play once and you either win or lose. Play ten thousand times and the outcome is no longer in doubt: you will have lost money, and by a predictable amount.

Problem — where it breaks or why it matters

Single outcomes are unpredictable; long-run averages are not. We need the number that long-run averages converge to, computed from the distribution rather than from data.

Worked Arithmetic — concrete numbers

The game. Win \$10 with probability 1/8, \$0 with probability 7/8:

$$\mathbb{E}[\text{winnings}] = 10 \times \tfrac{1}{8} + 0 \times \tfrac{7}{8} = 1.25.$$

Cost \$2, so the expected profit is $1.25 - 2 = -0.75$ per play. Over 10,000 plays, expect to lose about \$7,500.

One die.

$$\mathbb{E}[X] = \tfrac{1}{6}(1+2+3+4+5+6) = \tfrac{21}{6} = 3.5.$$

Note that 3.5 is not a possible outcome. The expectation is a balance point, not a prediction.

Two dice. By the same weighting,

$$\mathbb{E}[X] = \tfrac{1}{36}(2\cdot 1 + 3\cdot 2 + \cdots + 12\cdot 1) = \tfrac{252}{36} = 7.$$

Which is $3.5 + 3.5$ — expectation adds. ✓

Linearity, checked. Let $Y = 2X + 1$ for one die. Directly: Y takes values $3, 5, 7, 9, 11, 13$, averaging $\frac{48}{6} = 8$. By the rule: $2\mathbb{E}[X] + 1 = 2(3.5) + 1 = 8$. ✓

Formal Name — the mathematics

The **expectation** of a random variable is

$$\mathbb{E}[X] = \sum_x x\,p(x) \quad \text{(discrete)}, \qquad \mathbb{E}[X] = \int_{-\infty}^{\infty} x\,f(x)\,dx \quad \text{(continuous)}.$$

It is **linear**: for constants a, b and any X, Y,

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y],$$

which holds whether or not X and Y are independent.
For a function of X, $\mathbb{E}[g(X)] = \sum_x g(x)p(x)$. In general $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$.

Where the Analogy Breaks

Expectation requires the defining sum or integral to converge, and for heavy-tailed distributions it may not. If $\mathbb{P}(|X| > x) \sim x^{-\alpha}$ with $\alpha \le 1$, then $\mathbb{E}[X]$ simply does not exist — not "is large" or "is hard to estimate", but is undefined. Our markets have $\hat{\alpha}$ between 2.54 and 3.21, so the mean and variance are fine, but this is a property to be checked rather than assumed, and §6.12 shows which moments do fail.

Why It Matters Here — link to the thesis

Every loss function minimised in this thesis is an expectation estimated by a batch average. The WGAN critic loss $\mathbb{E}[D(x_{\text{real}})] - \mathbb{E}[D(x_{\text{fake}})]$ and the gradient penalty $\mathbb{E}[(\|\nabla D\|_2 - 1)^2]$ are both expectations, approximated by averaging over a batch of 64 or 128 windows.

6.8 Variance and Higher Moments

Situation — the everyday picture

Two investments both average 5% a year. One returns between 4% and 6% every year; the other alternates +40% and −30%. The average is a wholly inadequate description of the difference.

Problem — where it breaks or why it matters

Location without spread is half a summary. And for financial returns even spread is not enough: the shape of the extremes matters more than their typical size, so we need a ladder of increasingly fine descriptions — and a clear account of when each rung exists.

Worked Arithmetic — concrete numbers

One die. $\mathbb{E}[X] = 3.5$. Deviations and their squares:

x	1	2	3	4	5	6
$x - 3.5$	-2.5	-1.5	-0.5	0.5	1.5	2.5
$(x-3.5)^2$	6.25	2.25	0.25	0.25	2.25	6.25

$$\mathrm{Var}(X) = \tfrac{1}{6}(6.25 + 2.25 + 0.25 + 0.25 + 2.25 + 6.25) = \tfrac{17.5}{6} = 2.9167,$$

so the standard deviation is $\sqrt{2.9167} = 1.708$.

The shortcut. $\mathbb{E}[X^2] = \frac{1}{6}(1 + 4 + 9 + 16 + 25 + 36) = \frac{91}{6} = 15.1667$, and

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 15.1667 - 12.25 = 2.9167. \quad \checkmark$$

Why squares, not absolute values. Squaring makes variance add for independent variables, which absolute deviation does not. For two independent dice, $\mathrm{Var}(\mathrm{sum}) = 2.9167 + 2.9167 = 5.8333$.

Scaling. $\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X)$: doubling a return quadruples its variance but only doubles its standard deviation. This is why volatility is quoted as a standard deviation — it shares the units of the returns themselves.

Formal Name — the mathematics

The **variance** is

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2,$$

and the **standard deviation** is $\sigma = \sqrt{\mathrm{Var}(X)}$.
The **k-th moment** is $\mathbb{E}[X^k]$. Standardised third and fourth moments give

$$\text{Skewness} = \frac{\mathbb{E}[(X-\mu)^3]}{\sigma^3}, \qquad \text{Kurtosis} = \frac{\mathbb{E}[(X-\mu)^4]}{\sigma^4},$$

with excess kurtosis defined as kurtosis minus 3, so that a normal distribution scores 0.
Properties: $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$, and $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ when X and Y are independent.

Where the Analogy Breaks

Higher moments exist only when their integrals converge. For a power-law tail $\mathbb{P}(|X| > x) \sim x^{-\alpha}$, the k-th moment is finite only if $k < \alpha$. The BRICS markets in this project have $\hat{\alpha}$ between 2.54 and 3.21:

- $k = 2$ (variance): finite everywhere, since $2 < 2.54$;
- $k = 3$ (skewness): infinite in four of five markets;
- $k = 4$ (kurtosis): *infinite in all five.*

The population kurtosis of these return series does not exist. A sample kurtosis can always be computed, but it is estimating nothing.

Why It Matters Here — link to the thesis

This is why `kurtosis_diff` and `skewness_diff` were moved out of the ranked metrics and into the descriptive-only group. Ranking models on an estimate of a quantity that is infinite is not a strict test — it is noise with a decimal point, and Chapter 9 shows the estimate diverging with sample size rather than converging.

6.9 Joint Distributions and Independence

Situation — the everyday picture

Height and weight, measured across a population. Knowing someone is tall changes what you expect their weight to be. Height and shoe colour, by contrast, tell you nothing about each other.

Problem — where it breaks or why it matters

Individual distributions cannot express a relationship. Two return series may have identical histograms and behave completely differently in combination — and the whole question of temporal structure is a question about the joint behaviour of r_t and r_{t-1}.

Worked Arithmetic — concrete numbers

Two dice. Let X be the first roll and S the sum.

Are the two rolls independent? $\mathbb{P}(\text{first} = 3) = 1/6$ and $\mathbb{P}(\text{second} = 5) = 1/6$; the pair $(3, 5)$ is one of 36 outcomes, so

$$\mathbb{P}(3, 5) = \tfrac{1}{36} = \tfrac{1}{6} \times \tfrac{1}{6}. \quad \checkmark$$

The joint factorises, so they are independent.

Are X and S independent? $\mathbb{P}(X = 1) = 1/6$ and $\mathbb{P}(S = 12) = 1/36$. If independent, $\mathbb{P}(X = 1, S = 12)$ would be $\frac{1}{6} \times \frac{1}{36} = \frac{1}{216}$. But $S = 12$ requires both dice to be 6, so $X = 1$ and $S = 12$ cannot both happen:

$$\mathbb{P}(X = 1, S = 12) = 0 \neq \tfrac{1}{216}.$$

Not independent. ✓Knowing the sum tells you about the first roll.

Conditional distribution. Given $S = 4$, the possibilities are $(1, 3), (2, 2), (3, 1)$, so $X \in \{1, 2, 3\}$ each with probability $1/3$ — a completely different distribution from the unconditional uniform on $\{1, \dots, 6\}$.

Formal Name — the mathematics

The **joint distribution** of X and Y gives $\mathbb{P}(X = x, Y = y)$, or in the continuous case a joint density $f(x, y)$ with

$$\mathbb{P}((X, Y) \in A) = \iint_A f(x, y)\, dx\, dy.$$

The **marginal** distribution of X recovers X alone by summing or integrating out Y.
X and Y are **independent** if the joint factorises for all values:

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y), \qquad \text{or} \qquad f(x, y) = f_X(x) f_Y(y).$$

Equivalently, the conditional distribution of X given Y does not depend on Y.

Where the Analogy Breaks

Marginals do not determine the joint. Two pairs of variables can have identical individual distributions and utterly different dependence — this is exactly the gap a copula is designed to describe. It is also the precise sense in which matching a histogram is not matching a process, and the reason the evaluation framework in this thesis needs a second family of metrics at all.

Why It Matters Here — link to the thesis

The shuffled-real control has *identical* marginals to the real data by construction and destroyed joint structure. That single fact is why an evaluation built only on marginal distances ranked it first: under the old unweighted 19-metric composite it scored an average rank of 1.24 against 2.47 for a genuine generator. The permutation-invariant metrics could not see the difference, because there is no difference for them to see.

6.10 Covariance and Correlation

Situation — the everyday picture

Ice cream sales and drowning deaths rise together over the year. Neither causes the other — both follow temperature — but the co-movement is real and measurable.

Problem — where it breaks or why it matters

Independence is a yes/no property. What is usually wanted is a graded measure: *how strongly* do two quantities move together, on a scale that permits comparison across quantities with different units?

Worked Arithmetic — concrete numbers

Four paired observations:

x	1	2	3	4
y	2	4	5	9

Means: $\bar{x} = 10/4 = 2.5$, $\bar{y} = 20/4 = 5$.

$x - \bar{x}$	-1.5	-0.5	0.5	1.5
$y - \bar{y}$	-3	-1	0	4
product	4.5	0.5	0	6

$$\mathrm{Cov}(x, y) = \tfrac{1}{4}(4.5 + 0.5 + 0 + 6) = \tfrac{11}{4} = 2.75.$$

Positive: the two rise together.

To correlation. Spreads: $\mathrm{Var}(x) = \frac{1}{4}(2.25 + 0.25 + 0.25 + 2.25) = 1.25$, so $\sigma_x = 1.118$; $\mathrm{Var}(y) = \frac{1}{4}(9 + 1 + 0 + 16) = 6.5$, so $\sigma_y = 2.550$.

$$\rho = \frac{2.75}{1.118 \times 2.550} = \frac{2.75}{2.851} = 0.965.$$

Close to the maximum of 1 — a strong, nearly linear relationship.

Units cancel. Measure y in different units, say doubled: covariance doubles to 5.5, but σ_y doubles too, so ρ is unchanged. ✓

Formal Name — the mathematics

$$\mathrm{Cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y],$$

$$\rho(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} \in [-1,1].$$

$\rho = \pm 1$ exactly when Y is an increasing or decreasing linear function of X. If X and Y are independent then $\mathrm{Cov}(X,Y) = 0$.
The **autocorrelation** of a series at lag k is the correlation of the series with itself shifted by k:

$$\rho(k) = \frac{\mathrm{Cov}(r_t, r_{t-k})}{\mathrm{Var}(r_t)}.$$

Where the Analogy Breaks

Zero correlation does not imply independence. Let X be symmetric about zero and $Y = X^2$; then $\mathrm{Cov}(X,Y) = 0$ while Y is completely determined by X. Correlation detects *linear* association only.
This is exactly the situation for financial returns. The autocorrelation of r_t is near zero — returns are close to linearly unpredictable — while the autocorrelation of $|r_t|$ and r_t^2 is strongly positive and decays slowly. Returns are uncorrelated and emphatically not independent, which is the single most important structural fact in this thesis.

Why It Matters Here — link to the thesis

This is why the evaluation computes three separate autocorrelation metrics — `acf_returns_mae` on r_t, `acf_absolute_mae` on $|r_t|$, and `acf_squared_mae` on r_t^2. A generator that matches only the first has reproduced the absence of linear predictability while missing volatility clustering entirely, and only the second and third catch it.

6.11 The Law of Large Numbers

Situation — the everyday picture

Flip a fair coin ten times and you might see seven heads — unremarkable. Flip it ten thousand times and 7,000 heads would be extraordinary. The proportion settles toward one half, and it settles more firmly the longer you go.

Problem — where it breaks or why it matters

Expectation was defined from a distribution, and we asserted that long-run averages converge to it. That assertion needs to be a theorem, with stated conditions — because the conditions are exactly what financial data threatens.

Worked Arithmetic — concrete numbers

Rolling a fair die, $\mathbb{E}[X] = 3.5$. The standard deviation of the average of n rolls is $\sigma/\sqrt{n}$ with $\sigma = 1.708$:

n	sd of the average	typical spread
1	1.708	± 1.71
100	0.171	± 0.17
10,000	0.017	± 0.02

Each hundredfold increase in n shrinks the spread tenfold, since $\sqrt{100} = 10$. Convergence is real but slow: to halve the uncertainty you must quadruple the data.

The cost of that slowness. This project's per-market test set holds about 496 observations; pooling all five markets gives about 2,480. The pooled standard error is $\sqrt{2480/496} = \sqrt{5} = 2.24$ times smaller — which is why the downstream utility metrics are computed on the pooled set rather than per-market.

Formal Name — the mathematics

Law of Large Numbers. If $X_1, X_2, \ldots$ are independent with common distribution and $\mathbb{E}|X| < \infty$, then the sample mean

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \longrightarrow \mathbb{E}[X]$$

as $n \to \infty$. When the variance σ^2 is also finite, $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$, so the standard error is $\sigma/\sqrt{n}$.

Where the Analogy Breaks

The law requires the mean to exist. It says nothing about how fast convergence happens for any particular distribution, and for heavy-tailed data the answer can be "far more slowly than $1/\sqrt{n}$ suggests", because a single extreme observation can move the running average substantially even at large n. The guarantee is asymptotic; the sample you actually hold is finite.

Why It Matters Here — link to the thesis

Every metric in this thesis is an average over a finite sample and therefore carries a standard error. The discriminative AUC null is not exactly 0.5 but 0.506 ± 0.084 — measured across fifteen real-versus-real splits — and that spread is precisely the $\sigma/\sqrt{n}$ of this section. An observed AUC must be judged against that spread, not against the theoretical 0.5.

6.12 The Central Limit Theorem — and Where It Fails

Situation — the everyday picture

Roll a die 100 times and record the average. Repeat the whole experiment 10,000 times and plot the 10,000 averages. A bell curve appears — even though a single die is flat, with all six faces equally likely. The bell was not in the die; it came from the averaging.

Problem — where it breaks or why it matters

The Central Limit Theorem is the reason the normal distribution is everywhere, and it is routinely over-read as licensing Gaussian models for anything. Applied to daily market returns it makes predictions that are wrong by orders of magnitude, and understanding exactly which of its conditions fails — and which does not — is the intellectual core of this thesis.

Worked Arithmetic — concrete numbers

What the Gaussian predicts. For a normal distribution,

$$\mathbb{P}(|Z| > 5) = 2\Phi(-5) \approx 5.73 \times 10^{-7}.$$

At 252 trading days per year, the expected waiting time for one such day is

$$\frac{1}{252 \times 5.73 \times 10^{-7}} \approx 6{,}922 \text{ years.}$$

What the data shows. A single BRICS market contributes roughly 6,201 market-days — about 25 years. The Gaussian expects $6201 \times 5.73 \times 10^{-7} \approx 0.0036$ five-sigma days over that span: call it one day every three centuries. Eleven were observed.
The discrepancy is a factor of roughly $11/0.0036 \approx 3{,}000$. This is not a model that is slightly miscalibrated in the tails; it is a model whose tail predictions are meaningless.

Which condition failed? The CLT needs finite variance. Our markets have $\hat{\alpha} \in [2.54, 3.21] > 2$, so the variance *is* finite and the CLT *does* apply to averages of returns. The theorem is not violated.
What fails is the inference people draw from it. The CLT describes the distribution of the *average* of many returns. It says nothing whatever about the distribution of a *single* return, and a five-sigma day is a single observation, not an average. Modelling individual daily returns as Gaussian is not an application of the CLT — it is an unrelated assumption that happens to be false.

And the fourth moment. Even where the CLT applies, it applies to the mean. The analogous result for the sample variance requires a finite fourth moment, and with $\alpha < 4$ everywhere in our data that moment is infinite. So the sample kurtosis has no limiting normal distribution, no standard error, and no convergence — Chapter 9 shows it climbing from 2.53 at $n = 250$ to 11.84 at $n = 2{,}000$ on BOVESPA while the Hill estimator, which does not require finite moments, settles from 4.83 to 3.11.

Formal Name — the mathematics

Central Limit Theorem. Let $X_1, X_2, \ldots$ be independent with common distribution, mean μ and *finite* variance σ^2. Then

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \longrightarrow \mathcal{N}(0, 1)$$

in distribution as $n \to \infty$.
Three conditions carry the weight: independence, identical distribution, and finite variance. When the variance is infinite ($\alpha < 2$), averages converge instead to a **stable** distribution with heavy tails of their own, and the normal approximation never appears at any sample size.

Where the Analogy Breaks

The CLT is about sums and averages, not about individual observations, and this is the confusion it most often causes. It also assumes independence, which market returns violate through volatility clustering — versions of the theorem exist for dependent sequences, but they converge more slowly and require their own conditions.

So there are two distinct failures to keep apart. Applying the CLT to a single day's return is a misapplication of a correct theorem. Computing a sample kurtosis and treating it as an estimate is applying a theorem whose conditions genuinely do not hold.

Why It Matters Here — link to the thesis

This section is the motivation for the entire thesis. If daily returns were Gaussian, a closed-form risk model would suffice and no generative model would be needed. Because they are not — eleven five-sigma days where the model expects 0.0036 — we need generators that reproduce heavy tails and volatility clustering, and an evaluation framework honest enough to check whether they actually do. It is also why the tail index $\hat{\alpha}$, which needs no finite moments, is a ranked metric while kurtosis is not.

Chapter 7

Statistics

Before and After

Before this chapter you need: logarithms and the log rules (§2.3), derivatives and how to find a maximum (§3.3–§3.4), gradients (§4.3), random variables, expectation and variance (§6.5–§6.8), the law of large numbers and the CLT (§6.11–§6.12).
After it you will understand: why an estimate is itself a random quantity, what maximum likelihood maximises, why this project fits GARCH with a Gaussian likelihood it knows to be wrong, what a p-value does and does not mean, why a test on 125 days is nearly powerless, and what goes wrong when 19 metrics are tested at once.

Probability reasons from a known distribution to the behaviour of data. Statistics runs the arrow backwards: from data in hand to claims about the distribution that produced it. Every number reported in this thesis is a statistic, and every one carries uncertainty this chapter makes explicit.

7.1 Samples and Sampling Distributions

Situation — the everyday picture

You measure the temperature every day for a week: 15, 17, 14, 19, 16, 21, 13 degrees. You want the "typical" value and how much the values vary. Next week you repeat the exercise and get different numbers — and therefore a different average.

Problem — where it breaks or why it matters

The second week's average differs from the first, yet neither is wrong. The summary you compute is not a fixed property of the world but something that changes with the sample. Unless that variation is quantified, there is no way to tell a real difference from an accident of sampling.

Worked Arithmetic — concrete numbers

The week's data. $x = \{15, 17, 14, 19, 16, 21, 13\}$.

Mean: $\bar{x} = (15 + 17 + 14 + 19 + 16 + 21 + 13)/7 = 115/7 = 16.43$.
Deviations: $\{-1.43, 0.57, -2.43, 2.57, -0.43, 4.57, -3.43\}$.
Squared: $\{2.04, 0.33, 5.90, 6.60, 0.18, 20.88, 11.76\}$, totalling 47.69.
Sample variance: $s^2 = 47.69/6 = 7.95$, so $s = 2.82$°C.

Why divide by 6 and not 7. The deviations are taken from $\bar{x}$, which was computed from the same seven numbers. They therefore sum to zero by construction, so only six of them are free — fix any six and the seventh follows. Dividing by $n-1$ corrects for that, and without it the variance would be systematically too small.

The sampling distribution. If temperatures have true standard deviation $\sigma = 3$, then across repeated weeks the sample mean has standard deviation

$$\frac{\sigma}{\sqrt{n}} = \frac{3}{\sqrt{7}} = 1.13.$$

So a second week averaging 17.5 instead of 16.4 is entirely unremarkable: the gap of 1.1 is one standard error.

Formal Name — the mathematics

A **sample** is n observations drawn from a population. A **statistic** is any function of the sample; being a function of random quantities, it is itself a random variable, and its distribution across repeated samples is its **sampling distribution**.

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

The **standard error** of the mean is $\sigma/\sqrt{n}$, estimated by $s/\sqrt{n}$. The $n-1$ divisor is **Bessel's correction**.

Where the Analogy Breaks

Standard deviation summarises spread well only for roughly symmetric distributions. Returns fall much further than they rise, so a single number understates downside risk. And the $\sigma/\sqrt{n}$ standard error is itself a CLT result requiring finite variance — available for returns ($\hat{\alpha} > 2$) but not for their fourth moment, which is why no standard error is ever quoted for a sample kurtosis in this thesis.

Why It Matters Here — link to the thesis

Daily volatility $\sigma = \mathrm{sd}(r_t)$ is the foundational scale in all five markets: MOEX 1.87%/day, BOVESPA 1.63%. Annualising multiplies by $\sqrt{252}$. Every reported metric is a statistic with a sampling distribution, which is why the pooled test set of about 2,480 observations is preferred to the per-market 496 wherever the metric permits pooling.

7.2 Estimators: Bias and Variance

Situation — the everyday picture

Two bathroom scales. One reads exactly 2 kg heavy every time — consistent but wrong. The other averages correctly across many weighings but scatters ±3 kg. Neither is obviously better; it depends what you need.

Problem — where it breaks or why it matters

"Accurate" bundles together two distinct failures — being systematically off, and being erratic — and they trade against each other. Separating them is what lets you choose an estimator deliberately rather than by habit.

Worked Arithmetic — concrete numbers

True weight 70 kg. Five weighings on each scale.

	readings					mean	spread
Scale A	72	72	72	72	72	72	0
Scale B	67	73	70	68	72	70	2.55

Scale A: bias $= 72 - 70 = +2$, variance $= 0$.
Scale B: bias $= 0$, standard deviation $= 2.55$.

Which is better? Compare mean squared error, MSE $=$ bias^2 $+$ variance:

$$\text{A: } 2^2 + 0 = 4, \qquad \text{B: } 0^2 + 6.5 = 6.5.$$

The biased scale wins — on this criterion, at this sample size. Averaging four readings from B, however, cuts its variance to $6.5/4 = 1.63$ and its MSE to 1.63, beating A. The right choice depends on how much data you have.

Bessel's correction as bias removal. On the seven temperatures, dividing by $n = 7$ gives $47.69/7 = 6.81$; by $n - 1 = 6$ gives 7.95. The first is biased low; the second is unbiased.

Formal Name — the mathematics

An **estimator** $\hat{\theta}$ is a statistic used to estimate a parameter θ. Its **bias** is $\mathbb{E}[\hat{\theta}] - \theta$, and it is **unbiased** when this is zero.

$$\mathrm{MSE}(\hat{\theta}) = \mathbb{E}\big[(\hat{\theta} - \theta)^2\big] = (\text{bias})^2 + \mathrm{Var}(\hat{\theta}).$$

$\hat{\theta}$ is **consistent** if it converges to θ as $n \to \infty$. Consistency is about the limit; unbiasedness is about any fixed n. Neither implies the other.

Where the Analogy Breaks

Unbiasedness is a weaker virtue than its name suggests. It guarantees correctness on average over hypothetical repetitions, which is cold comfort when you have one sample. A biased estimator with much smaller variance is frequently the better instrument, and every regularised model in machine learning — including weight decay and early stopping — deliberately accepts bias to reduce variance.

Why It Matters Here — link to the thesis

The Hill estimator is biased at small samples yet consistent, and its bias shrinks as the tail sample grows. This is why Chapter 9 reports it converging from 4.83 at $n = 250$ to 3.11 at $n = 2{,}000$ on BOVESPA: the movement is bias disappearing, not instability. Sample kurtosis over the same range moves the opposite way — 2.53 to 11.84 — because it is not converging to anything at all.

7.3 Maximum Likelihood

Situation — the everyday picture

A coin lands heads 7 times in 10 flips. What is its bias? You would guess 0.7, and the reasoning behind that guess is: of all possible biases, 0.7 is the one that makes what actually happened most probable.

Problem — where it breaks or why it matters

That reasoning needs to become a procedure that works for any model, not just coins — one that turns "which parameter best explains this data" into a calculation.

Worked Arithmetic — concrete numbers

The coin. For bias p, the probability of a specific sequence with 7 heads in 10 flips is $p^7(1-p)^3$. Evaluate:

p	0.5	0.6	0.7	0.8	0.9
$p^7(1-p)^3$	0.000977	0.001791	**0.002224**	0.001678	0.000478

The largest value is at $p = 0.7$. ✓

By calculus. Maximise the logarithm instead — the log is increasing, so it has its maximum in the same place, and it turns the product into a sum (§2.3):

$$\ell(p) = 7\ln p + 3\ln(1-p).$$

Differentiate and set to zero, using $\frac{d}{dp}\ln p = 1/p$:

$$\ell'(p) = \frac{7}{p} - \frac{3}{1-p} = 0 \implies 7(1-p) = 3p \implies 7 = 10p \implies p = 0.7. \quad \checkmark$$

Confirm it is a maximum. $\ell''(p) = -7/p^2 - 3/(1-p)^2 < 0$ for every p in $(0,1)$, so the function is concave and 0.7 is its highest point (§3.4). ✓

Formal Name — the mathematics

Given data $x_1, \ldots, x_n$ and a model with density $f(x \mid \theta)$, the **likelihood** treats the data as fixed and θ as the argument:

$$L(\theta) = \prod_{i=1}^{n} f(x_i \mid \theta), \qquad \ell(\theta) = \ln L(\theta) = \sum_{i=1}^{n} \ln f(x_i \mid \theta).$$

The **maximum likelihood estimator** is $\hat{\theta} = \arg\max_\theta \ell(\theta)$.
Under regularity conditions — including that the model is correctly specified — MLE is consistent, asymptotically unbiased, and asymptotically normal with the smallest achievable variance.

Where the Analogy Breaks

The likelihood is not a probability distribution over θ. It is a function of θ computed from fixed data, and it does not integrate to one; reading $L(0.7) = 0.0022$ as "the probability the bias is 0.7" is a category error. Turning the likelihood into a distribution over θ requires a

prior and Bayes' theorem (§6.4).
The optimality guarantees also assume the model is right. When it is not, the next section describes what survives.

Why It Matters Here — link to the thesis

Log-likelihood is maximised whenever a GARCH model is fitted in this project. The sum form is not a convenience but a necessity: a product of several thousand small densities underflows to zero in floating-point arithmetic, while the corresponding sum of logarithms is perfectly stable.

7.4 QMLE: Consistency Under Misspecification

Situation — the everyday picture

You measure a room with a tape that stretches slightly under tension. Every individual reading is a little off. But if the stretch is proportional, the *ratio* of two walls is still exactly right — the flawed instrument gets some quantities wrong while getting others right.

Problem — where it breaks or why it matters

Maximum likelihood's guarantees assume the model is correct. Real models never are. The practical question is not whether the model is wrong — it is — but which of its outputs survive being wrong, and this project's residual-kurtosis test depends entirely on the answer.

Worked Arithmetic — concrete numbers

The setup. GARCH$(1,1)$ models the conditional variance:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \qquad \varepsilon_t = \sigma_t z_t,$$

where z_t is a standardised innovation. Fitting requires assuming a distribution for z_t.

The test we want to run. Real returns keep excess kurtosis *after* dividing out the time-varying volatility (Bollerslev 1987). That is stylised fact 7: fat tails are not fully explained by volatility clustering. We want to measure that leftover kurtosis in the standardised residuals $\hat{z}_t = \varepsilon_t/\hat{\sigma}_t$ and check whether a generator reproduces it.

Why not assume a Student-t? A t innovation would fit the data better — the returns really are heavy-tailed. But it fits better precisely by absorbing the excess kurtosis into the assumed innovation distribution. The residuals would then look t-shaped by construction, and measuring their kurtosis would recover the assumption rather than a property of the data. The test would be circular.

Why Gaussian works anyway. Maximising a Gaussian likelihood estimates (ω, α, β) consistently even when z_t is not Gaussian. The Gaussian score equations depend on the data only through first and second conditional moments, and those are correctly specified by the GARCH recursion whatever the shape of z_t. So:

- the variance path $\hat{\sigma}_t$ is estimated consistently — *right*;
- the standard errors from the Gaussian information matrix are — *wrong*, and need a sandwich correction;
- leftover kurtosis in $\hat{z}_t$ is — *genuine evidence*, because nothing in the fitting procedure put it there.

The deliberately wrong assumption is what makes the measurement honest.

Formal Name — the mathematics

Quasi-maximum likelihood estimation (QMLE) maximises a likelihood known to be misspecified. For GARCH, White (1982) and Bollerslev–Wooldridge (1992) establish that the Gaussian QMLE of the conditional-variance parameters is consistent and asymptotically normal under mild conditions, whatever the true innovation distribution.
The asymptotic variance is *not* the usual inverse information matrix but the **sandwich** form

$$A^{-1}BA^{-1},$$

where A is the expected Hessian and B the variance of the score. Under correct specification $A = B$ and this collapses to the standard result.

Where the Analogy Breaks

Consistency under misspecification is a guarantee about specific parameters, not a licence to ignore the model being wrong. Gaussian QMLE gets the variance dynamics right and the uncertainty quantification wrong. It also assumes the *variance equation* is correctly specified — if the true process has leverage effects or long-memory volatility that GARCH$(1,1)$ cannot represent, no choice of innovation distribution repairs that.

Why It Matters Here — link to the thesis

This is the reasoning behind `resid_kurtosis_diff`, one of the seven ranked temporal metrics. The pipeline fits a constant-mean GARCH$(1,1)$ by Gaussian QMLE, extracts $\hat{z}_t = r_t/\hat{\sigma}_t$, and compares the excess kurtosis of real and synthetic residuals. A generator that reproduces volatility clustering but not conditional fat tails will match the ACF metrics and fail this one — which is exactly the discrimination the metric exists to provide.

7.5 Hypothesis Testing

Situation — the everyday picture

A colleague claims they can tell tap water from bottled. You pour ten pairs and they get eight right. Impressive — or is it? Someone guessing would average five, and getting eight by luck alone is not extraordinary.

Problem — where it breaks or why it matters

Distinguishing a real effect from a lucky run needs a rule fixed in advance. Without one, any result can be narrated as impressive after the fact.

Worked Arithmetic — concrete numbers

Null hypothesis: they are guessing, so each trial is a fair coin, $p = 0.5$.
Probability of 8 or more correct out of 10 by chance:

$$\mathbb{P}(8) = \binom{10}{8}(0.5)^{10} = 45/1024 = 0.0439$$

$$\mathbb{P}(9) = \binom{10}{9}(0.5)^{10} = 10/1024 = 0.0098, \qquad \mathbb{P}(10) = 1/1024 = 0.0010$$

$$\mathbb{P}(\geq 8) = (45 + 10 + 1)/1024 = 56/1024 = 0.0547.$$

At the conventional 5% threshold, $0.0547 > 0.05$: do not reject. Eight out of ten is not quite enough.
Nine out of ten: $\mathbb{P}(\geq 9) = 11/1024 = 0.0107 < 0.05$. Reject.

The line between "unconvincing" and "convincing" falls between 8 and 9 correct — a single trial. That sharpness is an artefact of the threshold, not a feature of the evidence.

Formal Name — the mathematics

A test contrasts a **null hypothesis** H_0 with an **alternative** H_1. A **test statistic** is computed, and its distribution under H_0 determines whether the observed value is surprising. Two errors are possible:

- **Type I**: rejecting a true H_0. Probability α, chosen in advance, conventionally 0.05.
- **Type II**: failing to reject a false H_0. Probability β; the **power** is $1 - \beta$ (§7.8).

Failing to reject is not evidence for H_0. It means the data was insufficient to rule it out.

Where the Analogy Breaks

The 0.05 threshold is a convention with no mathematical standing — Fisher proposed it as a rough guide, not a law. Treating 0.049 and 0.051 as categorically different, when they differ by less than the sampling noise in either, is the single most common abuse of the framework.

Why It Matters Here — link to the thesis

The Kupiec test asks whether an observed VaR exceedance rate is distinguishable from the claimed 5%; Christoffersen asks whether exceedances cluster; the Anderson–Darling and Kolmogorov–Smirnov tests compare distributions. Each has a null, a statistic and a threshold — and §7.8 shows that on per-market samples several of them cannot reject anything.

7.6 What a p-Value Is and Is Not

Situation — the everyday picture

A study reports $p = 0.03$ and the write-up says "there is a 3% chance the result is due to chance". That sentence is wrong, and the correct statement is narrower and less satisfying than most readers want.

Problem — where it breaks or why it matters

The p-value is the most misread number in quantitative work. Since several appear in this thesis — Kupiec, Christoffersen, ARCH, Anderson–Darling — what each one licenses needs stating precisely.

Worked Arithmetic — concrete numbers

Return to eight correct out of ten. The p-value is 0.0547.

What it means: *if* the person were guessing, the probability of seeing eight or more correct is 0.0547.

What it does not mean:

- Not the probability they are guessing. That is $\mathbb{P}(H_0 \mid \text{data})$, and getting it from $\mathbb{P}(\text{data} \mid H_0)$ requires a prior and Bayes' theorem (§6.4) — the two differ by exactly the factor that made a positive disease test only 9% informative.
- Not the probability the result is a fluke.
- Not a measure of effect size. With $n = 10{,}000$ a trivial edge such as 50.5% produces a tiny p; with $n = 10$ a large edge produces a big one.
- Not comparable across tests as a strength ranking.

A concrete demonstration of the last point. Someone correct on 51% of 100,000 trials gives $p \approx 0.00003$ — overwhelming significance for an edge nobody would care about. Someone correct on 90% of 10 trials gives $p = 0.0107$. The second is the far more impressive ability and the less significant result.

Formal Name — the mathematics

The ***p*-value** is the probability, computed under H_0, of observing a test statistic at least as extreme as the one obtained:

$$p = \mathbb{P}(T \geq t_{\text{obs}} \mid H_0).$$

It is a statement about data given a hypothesis, never about a hypothesis given data. Under H_0 the p-value is uniformly distributed on $[0, 1]$ — which is why one test in twenty falls below 0.05 when nothing whatever is going on.

Where the Analogy Breaks

Some p-values in this thesis are not even usable as continuous evidence. `scipy`'s `anderson_ksamp` clamps its result to the range $[0.001, 0.25]$ and warns whenever the true value falls outside. A reported 0.001 may mean 0.001 or 10^{-12}; the number has been censored. The `arch_pvalue_diff` metric was moved to descriptive-only partly for this reason — it saturates on real data, giving $|\Delta p| = 0.000000$ and no discrimination at all.

Why It Matters Here — link to the thesis

Kupiec and Christoffersen p-values are reported per model. They are read as “the data does not contradict correct coverage”, never as “the model is correct” — and given the power calculations in the next section, on per-market samples the first statement is often true simply because almost nothing could contradict it.

7.7 Confidence Intervals

Situation — the everyday picture

A poll reports 52% support, "margin of error 3 points". The single number 52 is less informative than the range 49–55, because the range carries its own uncertainty with it.

Problem — where it breaks or why it matters

A point estimate invites false precision. Reporting $\hat{\alpha} = 3.11$ suggests a confidence that a sample of a few hundred tail observations cannot support, and an interval says so honestly.

Worked Arithmetic — concrete numbers

The seven temperatures: $\bar{x} = 16.43$, $s = 2.82$, $n = 7$.

Standard error: $s/\sqrt{n} = 2.82/\sqrt{7} = 1.066$.
95% interval (using ≈ 2 standard errors):

$$16.43 \pm 2 \times 1.066 = 16.43 \pm 2.13 = [14.30,\ 18.56].$$

The effect of more data. With the same s but $n = 700$:

$$\mathrm{SE} = 2.82/\sqrt{700} = 0.107, \qquad \text{interval} = 16.43 \pm 0.21.$$

A hundredfold increase in data narrows the interval tenfold — the $\sqrt{n}$ of §6.11 again.

The discriminative AUC null. Fifteen real-versus-real splits gave 0.506 ± 0.084. An observed AUC of 0.62 sits

$$z = \frac{0.62 - 0.506}{0.084} = 1.36$$

standard errors above the null centre, giving $p \approx 0.17$. Not significant — despite 0.62 looking comfortably above 0.5 to the naked eye.

Formal Name — the mathematics

A $(1 - \alpha)$ **confidence interval** is constructed so that, across repeated samples, the interval contains the true parameter a proportion $1 - \alpha$ of the time. For a mean with known σ:

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \qquad z_{0.025} = 1.96.$$

With σ estimated by s and small n, the t-distribution replaces the normal.

Where the Analogy Breaks

The confidence is in the procedure, not in the particular interval. Once computed, $[14.30, 18.56]$ either contains the true mean or it does not — there is no probability left. "95% probability the mean lies in this interval" is a Bayesian credible interval, which requires a prior and is a different object.

Note also that the interval above assumes the CLT applies. For a statistic with no finite variance — sample kurtosis on our data — the construction is unavailable in principle, not merely difficult.

Why It Matters Here — link to the thesis

The 0.506 ± 0.084 null is why the evaluation reports `discriminative_auc_absz`, the distance from the null centre measured in standard errors, rather than the raw AUC. Judging a generator against a theoretical 0.5 would flag ordinary sampling noise as a detectable difference.

7.8 Statistical Power

Situation — the everyday picture

You test a bathroom scale for a 200 g bias by weighing yourself twice. The test will not detect it — not because the bias is absent, but because two weighings cannot resolve 200 g against day-to-day variation. The instrument is too blunt for the question.

Problem — where it breaks or why it matters

A test that fails to reject tells you one of two things: the null is true, or your test was too weak to notice. These are entirely different conclusions, and only a power calculation separates them.

Worked Arithmetic — concrete numbers

Kupiec, this project's numbers. A VaR model claims 5% exceedances. Suppose the truth is 7%. Can the test detect the difference?

Test days n	Power at true $p = 7\%$	Setting
125	17%	5-year, per market
496	$\approx 56\%$	20-year, per market
2,480	$\approx 99\%$	20-year, pooled across 5 markets

Reading the top row. With 125 days, a genuinely broken model — exceeding its VaR 40% more often than advertised — goes undetected five times in six. A non-rejection carries essentially no information.

Why pooling helps so much. Exceedance counts scale with n while their standard deviation scales with $\sqrt{n}$, so the signal-to-noise ratio grows as $\sqrt{n}$. Going from 496 to 2,480 multiplies it by $\sqrt{5} = 2.24$, and that is enough to move power from a coin flip to near certainty.

The asymmetry this creates. At $n = 125$ a non-rejection is uninformative, while a rejection still means something — Type I error is controlled at 5% regardless of power. Low power damages only the negative result.

Formal Name — the mathematics

The **power** of a test is

$$1 - \beta = \mathbb{P}(\text{reject } H_0 \mid H_1 \text{ true}).$$

It rises with the sample size n, with the size of the true effect, and with the significance level α (at the cost of more false positives). It falls as the data becomes noisier.
Conventionally 80% power is treated as adequate. Below roughly 50%, a non-rejection should not be reported as support for the null.

Where the Analogy Breaks

Power is computed against a *specific* alternative. "The test has 56% power" is meaningless until the alternative is named — here, a true exceedance rate of 7% against a claimed 5%. Against a true rate of 6% the power is much lower; against 15%, much higher. Any power figure is an answer to a particular question.

Why It Matters Here — link to the thesis

This is the reason downstream utility is evaluated on the pooled test set of about 2,480 observations rather than per market. It is also why QLIKE is computed pooled: at per-market sample sizes the metric inverts, scoring Gaussian noise above real data. Reporting a per-market Kupiec non-rejection as evidence that a generator produces well-calibrated VaR would be a claim the data cannot support at 17% power.

7.9 Multiple Testing

Situation — the everyday picture

Twenty researchers each test a different useless drug. Each uses the 5% threshold. On average one of them finds a significant result and publishes it. The drug does nothing; the statistics were applied correctly; the finding is still spurious.

Problem — where it breaks or why it matters

This thesis computes many metrics per model per market. Every one is an opportunity for a chance result, and the probability that *at least one* misleads grows quickly with their number.

Worked Arithmetic — concrete numbers

Each test has a 5% false-positive rate under the null, so a 95% chance of behaving. For m independent tests, the probability that all behave is 0.95^m:

Tests m	$\mathbb{P}$(no false positive)	$\mathbb{P}$(at least one)
1	0.950	0.050
5	0.774	0.226
10	0.599	0.401
19	0.377	**0.623**

At the 19 metrics of the original composite, the chance of at least one spurious "significant" result is 62% — more likely than not, with every individual test correctly calibrated.

Bonferroni. To hold the overall rate at 5%, test each at $\alpha/m = 0.05/19 = 0.00263$. Then $0.99737^{19} = 0.951$, so the family-wise error rate is about 4.9%. ✓

The price. A threshold of 0.00263 instead of 0.05 demands far stronger evidence per test, which drives power down — and §7.8 already showed power was the binding constraint. Correcting for multiplicity and retaining power both require more data; there is no free version.

Formal Name — the mathematics

With m tests each at level α, the **family-wise error rate** is

$$\text{FWER} = 1 - (1 - \alpha)^m$$

under independence. The **Bonferroni correction** tests each at α/m, guaranteeing FWER $\leq \alpha$ whether or not the tests are independent.
Less conservative alternatives — Holm's step-down procedure, or Benjamini–Hochberg control of the false discovery rate — trade a weaker guarantee for more power.

Where the Analogy Breaks

Bonferroni assumes the worst case and is conservative when tests are correlated, which these emphatically are: `acf_absolute_mae` and `acf_squared_mae` measure nearly the same thing, and Wasserstein and energy distance both respond to the same distributional gaps. The effective number of independent tests is well below 19, so a strict Bonferroni over-corrects. Knowing the correction is too strong does not, however, license skipping it.

Why It Matters Here — link to the thesis

This is part of why the evaluation framework ranks models rather than accumulating significance tests. Ranks avoid the multiplicity problem: they make no distributional claim per metric and cannot generate false positives to correct for. The composite score is the mean of a fidelity rank and a temporal rank, and the taxonomy in Chapter 15 exists precisely so that seven permutation-invariant metrics cannot outvote seven ordering-sensitive ones — which is what let the shuffled control win the old unweighted 19-metric average at 1.24 against 2.47.

7.10 Linear Regression

Situation — the everyday picture

Plot exam score against hours studied. The points scatter but trend upward. You draw the straight line that best follows them, and use it to guess the score for someone who studied six hours.

Problem — where it breaks or why it matters

"Best" needs defining. Infinitely many lines pass near the points, and without a criterion the choice is arbitrary — and unrepeatable by anyone else.

Worked Arithmetic — concrete numbers

Four observations, hours x and score y:

x	1	2	3	4
y	2	4	5	9

These are the numbers from §6.10, where we found $\bar{x} = 2.5$, $\bar{y} = 5$, Cov $= 2.75$ and $\text{Var}(x) = 1.25$.

Slope:

$$\hat{\beta} = \frac{\mathrm{Cov}(x, y)}{\mathrm{Var}(x)} = \frac{2.75}{1.25} = 2.2.$$

Intercept: the line passes through $(\bar{x}, \bar{y})$, so

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = 5 - 2.2 \times 2.5 = -0.5.$$

Fitted line: $\hat{y} = -0.5 + 2.2x$.

Fitted values and residuals:

x	1	2	3	4
y	2	4	5	9
$\hat{y}$	1.7	3.9	6.1	8.3
residual	0.3	0.1	−1.1	0.7

Residuals sum to $0.3 + 0.1 - 1.1 + 0.7 = 0$, as they must when an intercept is fitted. ✓
Sum of squared residuals: $0.09 + 0.01 + 1.21 + 0.49 = 1.80$.

R^2. Total squared deviation of y about its mean: $9 + 1 + 0 + 16 = 26$. So

$$R^2 = 1 - \frac{1.80}{26} = 0.931.$$

The line accounts for 93% of the variation in scores.

Formal Name — the mathematics

Simple linear regression models

$$y_i = \alpha + \beta x_i + \varepsilon_i,$$

and **ordinary least squares** chooses α, β to minimise $\sum_i (y_i - \alpha - \beta x_i)^2$. Setting both partial derivatives (§4.2) to zero gives

$$\hat{\beta} = \frac{\mathrm{Cov}(x, y)}{\mathrm{Var}(x)}, \qquad \hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}.$$

The **coefficient of determination** is

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2},$$

the fraction of variance explained. In matrix form with several predictors, $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$ — which exists exactly when $X^\top X$ is invertible (§5.3).

Where the Analogy Breaks

R^2 never decreases when predictors are added, even meaningless ones, so it cannot be used to choose between models of different size. It also says nothing about whether a linear model was appropriate: data lying on a perfect parabola can produce a respectable R^2 from a straight line that is wrong everywhere. And regression measures association, not causation — the ice cream and drowning of §6.10 would fit beautifully.

Why It Matters Here — link to the thesis

R^2 is the diagnostic reported for the TimeGAN autoencoder: reconstruction R^2 above 0.8 is required before Phase 3 is allowed to proceed, because a lossy embedding means the adversarial stage trains on noise. The smoke run reached $+0.995$. Regression also underlies the ARCH test, which regresses squared residuals on their own lags, and the discriminative classifier, whose logistic form is regression with a non-linear link.

Chapter 8

Time Series Analysis

A **time series** is a sequence of measurements indexed by time: $r_1, r_2, \ldots, r_T$. Market returns form a time series. This chapter explains the tools we use to characterise their structure.

8.1 Stationarity

Situation — the everyday picture

The temperature in Barcelona fluctuates around 18°C year-round (ignoring seasons). The sea level has been rising 3mm/year for decades. The temperature process is *stationary* — its statistical properties do not change over time. Sea level is *non-stationary* — its mean is drifting.

Problem — where it breaks or why it matters

If a time series is non-stationary, any statistics we compute (mean, variance, correlations) depend on when we measured them, making them meaningless for predicting the future. All our models assume stationarity of log returns.

Formal Name — the mathematics

A process $\{X_t\}$ is **weakly stationary** if:

1. $\mathbb{E}[X_t] = \mu$ (constant mean)
2. $\text{Var}(X_t) = \sigma^2 < \infty$ (constant finite variance)
3. $\text{Cov}(X_t, X_{t+k}) = \gamma(k)$ (covariance depends only on lag k, not t)

Raw price levels P_t are non-stationary (trending). Log returns $r_t = \ln(P_t/P_{t-1})$ are approximately stationary.

Where the Analogy Breaks

"Stationarity" in the weak sense does not require the distribution to be normal, only that the first two moments are constant. BRICS markets have constant mean (near zero) and stable variance in the long run, but their higher moments (skewness, kurtosis) exhibit regime changes (e.g., the MOEX invasion event).

Why It Matters Here — link to the thesis

All evaluation metrics in this thesis assume stationarity: the Wasserstein distance, ACF-MAE, Hurst exponent, and discriminative AUC are all computed on the assumption that the distribution of r_t does not change between train and test. The temporal 80/10/10 split (no shuffling) is the correct design for non-stationary data.

8.2 Autocorrelation Function (ACF)

Situation — the everyday picture

You write down today's temperature and tomorrow's temperature for 365 days. You plot them as (x_t, x_{t+1}) pairs. Because temperature is persistent (a hot day is likely to be followed by another hot day), the scatter plot shows a positive slope. The *correlation* between today's and tomorrow's temperature is around $+0.8$.
Now do the same for today's and next week's temperature: the correlation is lower, maybe $+0.4$. After a month, it might be nearly zero — last month's weather tells you little about today.

Problem — where it breaks or why it matters

For market returns, the correlation between r_t and r_{t+1} is *near zero* for all lags (fact 2: markets are approximately unpredictable). But the correlation between $|r_t|$ and $|r_{t+1}|$ — the absolute return, a measure of how wild each day was — is *significantly positive* for lags of 1 to 100 or more (volatility clustering).

Worked Arithmetic — concrete numbers

BOVESPA 20-year data ($n = 4{,}953$):

Lag k	$\hat{\rho}(r, k)$	$\hat{\rho}(\|r\|, k)$	$\hat{\rho}(r^2, k)$
1	$+0.03$	$+0.19$	$+0.17$
5	-0.01	$+0.14$	$+0.12$
20	$+0.01$	$+0.09$	$+0.08$
50	≈ 0	$+0.05$	$+0.04$

The raw return autocorrelation is near zero (unpredictable direction). The absolute and squared return autocorrelation is positive and decays slowly (persistent volatility). A good generator must produce synthetic $|r_t|$ sequences with the *same* slow decay.

Formal Name — the mathematics

The **sample autocorrelation** at lag k is:

$$\hat{\rho}(k) = \frac{\sum_{t=1}^{T-k}(x_t - \bar{x})(x_{t+k} - \bar{x})}{\sum_{t=1}^{T}(x_t - \bar{x})^2}$$

The **Autocorrelation Function** (ACF) is the sequence $\{\hat{\rho}(0), \hat{\rho}(1), \hat{\rho}(2), \ldots\}$.
Our metric **ACF-MAE** compares real and synthetic ACF vectors:

$$\text{ACF-MAE}(f) = \frac{1}{K}\sum_{k=1}^{K} |\hat{\rho}_{\text{real}}(k) - \hat{\rho}_{\text{syn}}(k)|, \quad K = 20$$

We compute this for three transforms: raw returns r_t, absolute returns $|r_t|$, squared returns r_t^2.

Where the Analogy Breaks

ACF-MAE measures average discrepancy, which can miss systematic biases at specific lags. However, it has the crucial property of being *continuous with no saturation*: unlike ARCH-LM, it does not underflow to 0 when applied to real fat-tailed data.

Why It Matters Here — link to the thesis

ACF-MAE on $|r_t|$ and r_t^2 is the *primary* ranking signal for volatility clustering (Temporal family). The shuffled control scores 0.043 on ACF(returns) and 0.076 on ACF($|r|$) — correctly penalised. Any generator that correctly reproduces clustering will have smaller ACF-MAE than the shuffled control on these metrics.

8.3 GARCH: The Volatility Clustering Model

Situation — the everyday picture

Think of the stock market as a mood thermometer. On quiet days, it reads "calm" and barely moves. On anxious days, it reads "stormy" and moves a lot. The key insight: the mood today is heavily influenced by the mood yesterday. If yesterday was stormy, today is probably also stormy.

Problem — where it breaks or why it matters

The normal distribution assigns the same probability to a 2% move on every day. Real markets have quiet periods where 0.5% moves are normal and turbulent periods where 2% moves are common. A model that ignores this will badly misjudge risk on quiet days (underestimating tail risk) and turbulent days (overestimating) simultaneously.

Worked Arithmetic — concrete numbers

GARCH(1,1) fit to BOVESPA (20-year):

$\omega = 0.000002$, $\alpha = 0.08$, $\beta = 0.91$
Long-run variance: $\sigma_\infty^2 = \omega/(1-\alpha-\beta) = 0.000002/(1-0.08-0.91) = 0.0002$
Long-run volatility: $\sigma_\infty = \sqrt{0.0002} \approx 1.41\%/\text{day}$ ✓
Persistence: $\alpha + \beta = 0.99$ (close to 1 = very slow mean-reversion)
On a day following a 3% return ($r_{t-1}^2 = 0.0009$):

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta\sigma_{t-1}^2 = 0.000002 + 0.08 \times 0.0009 + 0.91 \times 0.0002 = 0.000274$$

$\sigma_t = 1.66\%/\text{day}$ — 18% higher than the long-run average. Turbulence persists.

Formal Name — the mathematics

The **GARCH(1,1)** model (Bollerslev 1986) specifies:

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \overset{iid}{\sim} F(0,1)$$
$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

Constraints: $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$, $\alpha + \beta < 1$ (covariance-stationarity). Persistence $= \alpha + \beta$ (close to 1 in BRICS markets: slow volatility mean-reversion).
Gaussian QMLE: we fit assuming $z_t \sim \mathcal{N}(0,1)$ even when z_t has heavy tails. This gives consistent estimates of (ω, α, β) regardless of the true distribution. The residuals $\hat{\varepsilon}_t/\hat{\sigma}_t$ will exhibit excess kurtosis if the true conditional distribution is non-Gaussian (stylized fact 7).

Where the Analogy Breaks

GARCH is a parametric model. It assumes a specific mathematical form for how today's variance relates to yesterday's. Real markets show leverage effects (negative returns increase volatility more than positive returns of the same size), jump components, and long-memory volatility that GARCH(1,1) cannot fully capture. GARCH is a diagnostic tool here, not a generative model.

Why It Matters Here — link to the thesis

GARCH appears in two places: (1) **Conditional heavy-tail metric** — we fit Gaussian QMLE, extract residuals $\varepsilon_t/\hat{\sigma}_t$, and measure residual excess kurtosis; a good generator's synthetic residuals should have similar kurtosis. (2) **Downstream utility (TSTR)** — we fit GARCH on synthetic data, use the estimated parameters to filter real data, and compute conditional VaR backtest results.

8.4 ARCH Test

Situation — the everyday picture

You want to know whether a return series has volatility clustering or not. The ARCH-LM test gives a statistical answer: it checks whether squared past returns predict current squared returns.

Worked Arithmetic — concrete numbers

ARCH-LM regression: regress r_t^2 on $r_{t-1}^2, \ldots, r_{t-p}^2$ and test R^2.
LM statistic $= n \cdot R^2 \sim \chi^2(p)$ under the null of no ARCH.
Simulated GARCH data ($n = 1{,}200$): LM $\approx$ 56–125, p-value ≈ 0.0.
Real BRICS data (real series): LM ≈ 51, p-value $= 0.0$.
Real BRICS data (shuffled series): LM ≈ 67, p-value $= 0.0$.
The test saturates: both real and shuffled series give $p = 0.0$. The p-value difference $|\Delta p| = 0.000000$. The ARCH-LM p-value cannot distinguish a good generator from a shuffled deck on real BRICS data.

Formal Name — the mathematics

Engle's (1982) ARCH-LM statistic for p lags:

$$LM = n \cdot R^2_{\text{aux}}, \quad \text{where aux: } r_t^2 = \gamma_0 + \sum_{j=1}^{p} \gamma_j r_{t-j}^2 + \eta_t$$

Under H_0 (no ARCH): $LM \xrightarrow{d} \chi^2(p)$.

Where the Analogy Breaks

The ARCH-LM test is powerful in theory and in simulation (99% correct on simulated GARCH). It fails on real fat-tailed data because the squared-return regression is dominated by extreme values. Both real and shuffled series have the same squared-return distribution, so both produce large LM statistics and zero p-values. The test is correctly described as "saturated on real data".

Why It Matters Here — link to the thesis

Both `arch_pvalue_diff` and `arch_stat_diff` are placed in **DESCRIPTIVE_COLS** — computed and displayed for diagnostic transparency, but excluded from all composite ranking scores. ACF-MAE on $|r_t|$ is the primary volatility-clustering ranking signal because it never saturates.

8.5 Hurst Exponent and Long Memory

Situation — the everyday picture

The Nile River's floods are not independent year to year. A decade of high floods tends to be followed by another decade of high floods. This is *long memory*: correlations that decay so slowly they never fully disappear.

Worked Arithmetic — concrete numbers

For a process with long memory, the R/S statistic (range divided by standard deviation) scales as:

$$\mathbb{E}[R_n/S_n] \approx c \cdot n^H$$

For independent data: $H = 0.5$ exactly. For BRICS $|r_t|$ (volatility proxy): $H \approx 0.57$–0.65 typically.
Estimation at BOVESPA ($n = 4{,}953$): divide series into m blocks of size n/m, compute R/S for each, regress $\log(R/S)$ vs $\log(n)$, slope $= \hat{H}$.
Standard deviation: ≈ 0.022 at $n = 1{,}000$; ≈ 0.004 at $n = 4{,}000$. At 5-year data ($n = 995$): $\hat{H} \pm 0.022$ — barely significant above $H = 0.5$. At 20-year data ($n = 4{,}960$): $\hat{H} \pm 0.004$ — clearly significant.

Formal Name — the mathematics

The **Hurst exponent** $H \in (0, 1)$ characterises long-range dependence:

- $H = 0.5$: independent (random walk, Brownian motion)
- $H > 0.5$: positively correlated at long lags (persistent, long memory)

- $H < 0.5$: negatively correlated at long lags (anti-persistent, mean-reverting)

Applied to $|r_t|$, $H > 0.5$ implies that large-volatility periods have a tendency to persist longer than an independent process would predict (fact 4: long memory in volatility).

Where the Analogy Breaks

The R/S estimator is biased in small samples and sensitive to non-stationarities. The 20-year data substantially reduces this bias. We apply H to $|r_t|$, not r_t itself, because raw returns have $H \approx 0.5$ (the near-zero ACF fact) while volatility has $H > 0.5$.

Why It Matters Here — link to the thesis

`hurst_diff` $= |H_{\text{real}} - H_{\text{syn}}|$ is in the **Temporal** family. The shuffled control achieves Hurst diff ≈ 0.052 (correctly penalised: shuffling destroys long memory). A good generator should reproduce $H \approx 0.6$ for the absolute return series of each market.

Chapter 9

Heavy Tails and Power Laws

9.1 Power Laws

Situation — the everyday picture

City sizes: New York has 8 million people. The next three cities have 4M, 3M, 2M. Cut the population threshold in half, and twice as many cities qualify. This doubling relationship is a *power law*: the number of cities with population $> x$ scales like $x^{-\alpha}$ for some exponent α.

Problem — where it breaks or why it matters

The normal distribution has *exponential* tails: the probability of a 10σ event is not just much smaller than a 5σ event — it is impossibly smaller (ratio $\approx 10^{17}$). Power-law tails decay much more slowly, making extreme events genuinely probable on financial timescales.

Worked Arithmetic — concrete numbers

NIFTY50 daily returns (20-year, $n = 4{,}954$):
The top 5% of $|r_t|$ values: 248 observations, smallest $= 1.97\%$, largest $= 15.91\%$.
Hill estimate: $\hat{\alpha} = k / \sum_{i=1}^{k} \ln(|r|_{(n-i+1)}/|r|_{(n-k)})$
With $k = \lfloor 0.05n \rfloor = 248$: $\hat{\alpha} \approx 2.8$.
Implication: $\mathbb{P}(|r| > x) \approx C \cdot x^{-2.8}$.
At $x = 5\sigma$: normal predicts 5.7×10^{-7}; power-law predicts $\approx (5)^{-2.8} \times C$.
If $C = 0.05 \times 5^{2.8} \approx 3.3$: $\mathbb{P} \approx 3.3 \times 5^{-2.8} \approx 0.022 = 2.2\%$ vs Gaussian's 0.00006%.

Formal Name — the mathematics

A random variable X has a **power-law tail** with index $\alpha > 0$ if:

$$\mathbb{P}(|X| > x) \sim L(x) \cdot x^{-\alpha} \quad \text{as } x \to \infty$$

where $L(x)$ is a slowly varying function (converges to a constant). The **tail index** α determines which moments exist:

$$\mathbb{E}[|X|^k] < \infty \iff k < \alpha$$

With BRICS $\hat{\alpha} = 2.54$–3.21: variance exists ($k = 2 < \alpha$), skewness is borderline ($k = 3$: exists only for $\hat{\alpha} > 3$, i.e., only BOVESPA marginally), kurtosis does not exist anywhere ($k = 4 > \hat{\alpha}$ everywhere).

Where the Analogy Breaks

"Tail index" is an asymptotic property — it describes what happens in the extreme tail, not the bulk of the distribution. In finite samples (even 20-year daily data), the Hill estimator averages over both the power-law region and the Gaussian-like body of the distribution. The estimate is sensitive to the choice of k (number of order statistics used).

Why It Matters Here — link to the thesis

`tail_index_diff` $= |\hat{\alpha}_{\text{real}} - \hat{\alpha}_{\text{syn}}|$ is the primary heavy-tail metric in the **Fidelity** family. This replaces `kurtosis_diff`, which is undefined for $\hat{\alpha} < 4$. The Hill estimator has sd ≈ 0.035 at $n = 4{,}960$ — precise enough to distinguish generators that reproduce the tail index from those that do not.

9.2 Why Kurtosis Is Unreliable for BRICS Markets

Situation — the everyday picture

You want to measure how "extreme" a dataset is. The obvious measure is kurtosis: the fourth standardised moment. A larger kurtosis means more extreme events. You measure kurtosis on 250 days of BOVESPA data. Then on 2,000 days. The number changes dramatically.

Worked Arithmetic — concrete numbers

BOVESPA kurtosis at different sample sizes (demonstrated in the session):

Sample n	Sample kurtosis	Hill $\hat{\alpha}$
250	2.53	4.83 (sd 1.253)
2000	11.84	3.11 (sd 0.316)

Sample kurtosis *increases* as n grows — diverging rather than converging. Hill $\hat{\alpha}$ *converges* as n grows.

Mathematical reason: if $\mathbb{E}[X^4] = \infty$ (which occurs when $\hat{\alpha} < 4$), the sample kurtosis $\hat{\kappa}_n = \frac{n^{-1}\sum x_i^4}{(n^{-1}\sum x_i^2)^2}$ diverges almost surely as $n \to \infty$.

Formal Name — the mathematics

For a distribution with tail index α:

$$\mathbb{E}[|X|^k] < \infty \iff k < \alpha$$

When $\alpha < 4$ (as in all BRICS markets): $\mathbb{E}[X^4] = \infty$, so the population kurtosis is undefined. The **Law of Large Numbers** requires finite expectation; without it, the sample mean of X^4 does not converge. The sample kurtosis $\hat{\kappa}_n$ grows without bound, following a stable distribution with index $\alpha/4 < 1$.

Contrast with Hill: the Hill estimator uses log spacings of order statistics, which are $\Gamma(1,1)$ distributed under the power-law model. These have finite moments of all orders, making the Hill estimator consistent and asymptotically normal regardless of α.

Where the Analogy Breaks

"Sample kurtosis = 65.70 for MOEX" is not a measurement of the population kurtosis — the population kurtosis is infinite. It is an artifact of sample size, driven by the handful of extreme observations (especially 2022-02-24: −33.3%). If we took a different 20-year window, the sample kurtosis might be 30 or 150. The Hill estimate $\hat{\alpha} \approx 2.7$ for MOEX is comparatively stable.

Why It Matters Here — link to the thesis

This is the core motivation for TASK 17 in the pipeline. Kurtosis_diff and skewness_diff are demoted from FIDELITY_COLS (7 metrics, ranked) to DESCRIPTIVE_COLS (7 metrics, displayed but not ranked). tail_index_diff takes their role as the primary heavy-tail fidelity metric. The length guard in `FinancialMetrics.__init__` warns when comparing series of different lengths, because kurtosis differences between unequal-length series are biased by length alone.

9.3 Wasserstein Distance

Situation — the everyday picture

Imagine two piles of sand on a table. One pile represents the distribution of real returns; the other represents synthetic returns. You want to measure how different they are. The Wasserstein distance asks: what is the minimum amount of "work" (mass times distance) needed to move one pile into the shape of the other?

Problem — where it breaks or why it matters

The Kolmogorov-Smirnov (KS) test measures the maximum gap between two CDFs. It was designed for independent data. Market returns are not independent (ACF of $|r_t| > 0$ for 100+ lags). For correlated data, the effective sample size is $n_{\text{eff}} = n/(1 + 2\sum_k \rho_k) \ll n$. KS uses the nominal n, making its p-values severely anti-conservative.

Worked Arithmetic — concrete numbers

For one-dimensional distributions, the Wasserstein-1 distance is the L_1 distance between quantile functions:

$$W_1(P, Q) = \int_0^1 \left| F_P^{-1}(u) - F_Q^{-1}(u) \right| \, du$$

Practical computation: sort both series in ascending order. Pair each real return with its rank-matched synthetic return. Average absolute difference.
For the shuffled control: $W_1 = 0.000e{+}00$ (exact zero). The shuffled series is a permutation of the real data; sorted, they are identical.
For a well-trained generator: $W_1 \approx 0.001$–0.005 (residual fitting error).

Formal Name — the mathematics

The **Wasserstein-p distance** between distributions P and Q:

$$W_p(P, Q) = \left(\inf_{\gamma} \mathbb{E}_{(X,Y)\sim\gamma} \|X - Y\|^p \right)^{1/p}$$

where the infimum is over joint distributions (couplings) with marginals P and Q. For $p = 1$ in 1D: reduces to the integral of the absolute difference of CDFs:

$$W_1(P, Q) = \int_{-\infty}^{\infty} |F_P(x) - F_Q(x)| \, dx$$

Where the Analogy Breaks

Wasserstein is permutation-invariant: $W_1(\text{real}, \text{shuffled}) = 0$ because sorting both produces the same sequence. This is a feature (it correctly measures distributional proximity) and a limitation (it does not measure temporal proximity). This is precisely why it sits in FIDELITY rather than TEMPORAL.

Why It Matters Here — link to the thesis

`wasserstein` is one of the 7 Fidelity metrics. The shuffled-control audit revealed that Wasserstein, like all Fidelity metrics, gives a perfect score to the shuffled control. This motivated the three-score composite design: Fidelity metrics measure "do the individual values look right?"; Temporal metrics measure "does the ordering look right?".

Chapter 13

Generative Adversarial Networks

13.1 The GAN Idea

Situation — the everyday picture

A counterfeiter tries to make fake currency. A bank teller learns to detect fakes. Each tries to fool or catch the other. Over time, both get better — the counterfeiter produces more convincing fakes, the teller develops sharper detection skills. Eventually, the fakes may be indistinguishable from real currency.

Problem — where it breaks or why it matters

We want a machine that generates realistic market return sequences. A single neural network trained directly is hard to evaluate: how do you define a loss function that measures "realism" for a complex, high-dimensional sequence? The GAN answer: train a second network to play the role of the expert bank teller. It provides the loss signal automatically.

Formal Name — the mathematics

A **GAN** (Goodfellow et al., 2014) consists of two neural networks:

- **Generator** $G : \mathcal{Z} \to \mathcal{X}$: maps random noise $z \sim p_z$ to a synthetic sample $\tilde{x} = G(z)$
- **Discriminator** $D : \mathcal{X} \to [0, 1]$: classifies inputs as real ($D(x) = 1$) or synthetic ($D(\tilde{x}) = 0$)

Training is a minimax game:

$$\min_G \max_D \ \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z\sim p_z}[\log(1 - D(G(z)))]$$

At the Nash equilibrium: $p_G = p_{\text{data}}$ and $D(x) = 1/2$ everywhere.

Where the Analogy Breaks

The Nash equilibrium is a theoretical ideal. In practice, GAN training is unstable: mode collapse (generator produces only a few outputs), discriminator dominance (generator gradients vanish), and oscillation without convergence are common. WGAN-GP addresses the most severe of these issues.

Why It Matters Here — link to the thesis

All three generators in this thesis (TimeGAN, QuantGAN, FinGAN) are GANs. The GAN framework is chosen because it implicitly defines the target distribution through adversarial training rather than requiring an explicit density. This is essential for market returns, whose true density is unknown and non-Gaussian.

13.2 WGAN and the Wasserstein Distance

Situation — the everyday picture

The original GAN discriminator outputs a probability $\in [0, 1]$. When the generator's distribution is far from the real one, the discriminator saturates: it assigns near-zero probability to fake samples, and the gradient of $\log(1 - D(G(z)))$ vanishes. The generator cannot learn which direction to improve.

Problem — where it breaks or why it matters

Vanishing gradients in early training. Mode collapse (generator produces a single realistic example to fool the fixed discriminator). Training instability. These problems worsen when the real and generated distributions do not overlap — which is always the case early in training.

Formal Name — the mathematics

Wasserstein GAN (Arjovsky et al., 2017) replaces the discriminator with a **critic** $f : \mathcal{X} \to \mathbb{R}$ (no sigmoid) and minimises the Wasserstein-1 distance:

$$W_1(p_{\text{data}}, p_G) = \sup_{\|f\|_L \leq 1} \mathbb{E}_{x \sim p_{\text{data}}}[f(x)] - \mathbb{E}_{z \sim p_z}[f(G(z))]$$

by the Kantorovich-Rubinstein duality, where the supremum is over 1-Lipschitz functions.
WGAN-GP (Gulrajani et al., 2017) enforces the Lipschitz constraint via a gradient penalty:

$$\mathcal{L}_{GP} = \lambda \cdot \mathbb{E}_{\hat{x} \sim p_{\hat{x}}} \left[(\|\nabla_{\hat{x}} f(\hat{x})\|_2 - 1)^2 \right]$$

where $\hat{x} = \varepsilon x_{\text{real}} + (1 - \varepsilon) x_{\text{fake}}$, $\varepsilon \sim U[0, 1]$. Standard values: $\lambda = 10$, $n_{\text{critic}} = 5$.
Why $n_{\text{critic}} = 5$: the critic must accurately estimate W_1 before providing a useful gradient to the generator. Fewer updates underfit the critic, giving a biased gradient signal. 5 is the empirically validated minimum (Gulrajani et al., 2017).
Why $\beta_1 = 0$ in Adam: in adversarial training, the correct gradient direction for the generator reverses each time the critic is updated. Momentum ($\beta_1 > 0$) accumulates the previous direction and corrupts the current gradient. $\beta_1 = 0$ uses only the current gradient.

Where the Analogy Breaks

The gradient penalty is computed at interpolated points, not directly on training data. This means the Lipschitz constraint is only approximately enforced (not everywhere). In practice, WGAN-GP is far more stable than weight clipping (Arjovsky 2017) or spectral normalisation, and remains the standard.

Why It Matters Here — link to the thesis

Both QuantGAN and FinGAN use WGAN-GP with $n_{\text{critic}} = 5$, $\lambda_{gp} = 10$, Adam $\beta_1 = 0$, $\beta_2 = 0.9$. These hyperparameters are floors, not arbitrary choices — using $n_{\text{critic}} = 1$ or $\lambda_{gp} = 1$ would invalidate the Wasserstein approximation. This is documented in ED Table 3 of the paper.

13.3 TimeGAN

Situation — the everyday picture

You want to teach a machine to write convincing diary entries. First, you teach it to encode each entry into a compressed representation and decode back (autoencoder). Then you teach it to predict the next entry from the previous ones (supervisor). Only after these skills are established do you introduce the adversarial game.

Formal Name — the mathematics

TimeGAN (Yoon et al., 2019) has four components:

- **Embedder** $e : \mathcal{X} \to \mathcal{H}$: encodes real sequences to latent space
- **Recovery** $\hat{r} : \mathcal{H} \to \mathcal{X}$: decodes latent back to sequence
- **Supervisor** $s : \mathcal{H} \to \hat{\mathcal{H}}$: predicts next latent step from current
- **Generator** $G : \mathcal{Z} \to \mathcal{H}$: maps noise to latent sequences

Four training phases:

1. **Phase 1 — Autoencoder:** $\min \|X - \hat{r}(e(X))\|^2$
2. **Phase 2 — Supervisor:** $\min \|H_{t+1} - s(H_t)\|^2$ (temporal causality in latent space)
3. **Phase 3 — Joint adversarial:**

$$\mathcal{L}_G = \mathcal{L}_U + 100\sqrt{\mathcal{L}_S} + 100\,\mathcal{L}_V$$

   where $\mathcal{L}_U$ = unsupervised GAN loss, $\mathcal{L}_S$ = supervised latent loss, $\mathcal{L}_V$ = moment-matching on mean/std
4. **Phase 4 — Fine-tuning:** refine recovery with generated sequences

Backbone: GRU (Gated Recurrent Unit). Hidden dim = 24, 3 layers, lr = 10^{-3}.

Where the Analogy Breaks

The latent-space supervisor is the key innovation: by training the supervisor to predict the next latent state, TimeGAN forces the latent space to encode temporal causality. However, this requires many gradient steps per phase to work. With `epochs=5` (as in the current smoke-test pipeline), each phase gets approximately 2 gradient steps — far too few for the GRU to form any useful representation. TimeGAN results in the current draft should be interpreted as lower bounds.

Why It Matters Here — link to the thesis

TimeGAN's poor temporal ranking in the current results reflects training budget, not architectural weakness. The recommended training is $\geq 5{,}000$ iterations per phase (or ≥ 50 epochs at the current batch size). The RunPod production run will use adequate epochs; the current paper's TimeGAN results are explicitly flagged as undertrained.

13.4 QuantGAN

Situation — the everyday picture

Instead of processing a sequence step-by-step (like a recurrent network), imagine looking at the whole sequence at once with multiple "zoom levels": a short zoom sees the last 2 days, a medium zoom sees the last week, a long zoom sees the last month. The network combines all zoom levels simultaneously to capture structure at every timescale.

Formal Name — the mathematics

QuantGAN (Wiese et al., 2020) uses a **Temporal Convolutional Network** (TCN) with *causal dilated convolutions*:

- **Causal:** the convolution at time t only sees inputs at times $\leq t$ (no future leakage)
- **Dilated:** dilation d means the filter sees steps $t, t-d, t-2d, \dots$ (stride-d subsampling in the receptive field)

With K residual blocks of dilations $1, 2, 4, \dots, 2^{K-1}$, the receptive field $= 2^K$ steps.
Three TCN blocks with dilations 1/2/4: receptive field = 8 steps. For a 128-step sequence, this covers the full window.
WGAN-GP with $n_{\text{critic}} = 5$, $\lambda_{gp} = 10$. Noise dim = 100, lr = 10^{-4}.

Where the Analogy Breaks

The parallel multi-scale processing is the key advantage over RNNs: a GRU must "remember" information across many steps using a hidden state of fixed size, leading to gradient issues for long horizons. A dilated TCN accesses all timescales in one forward pass without vanishing gradients. The tradeoff: TCN is less natural for autoregressive modelling (each step predicts the next), and its inductive bias may over-smooth long-range structure.

Why It Matters Here — link to the thesis

QuantGAN's TCN architecture is well-suited to the multi-scale structure of BRICS volatility: intraday effects (1–5 days), weekly patterns (5 days), and monthly cycles (20 days) are all covered by dilations 1/2/4 in a 128-step window. QuantGAN achieves the best composite ranking in current (smoke-test) results.

13.5 FinGAN

Situation — the everyday picture

Image-generating GANs start from a small noisy image (4×4 pixels) and progressively upsample it to full resolution (256×256), adding detail at each step. FinGAN adapts this idea to sequences: start from a short noise vector and upsample via transposed convolutions to a full 128-step return sequence.

Formal Name — the mathematics

FinGAN (this paper): CNN deconvolution WGAN-GP.

- Generator: $3\times$ `ConvTranspose1d` with stride 2, base channels = 64
- Critic: mirror architecture with `Conv1d` and LayerNorm (not BatchNorm)

- seq_len must be divisible by 8 ($= 2^3$ from 3 upsampling blocks)

Why LayerNorm in the critic, not BatchNorm: the WGAN-GP gradient penalty is computed at a single interpolated point $\hat{x}$. BatchNorm normalises across the batch, introducing a dependency on other batch members that corrupts the per-point gradient norm calculation. LayerNorm normalises across channels at each point independently.

Where the Analogy Breaks

The image-GAN inductive bias assumes hierarchical structure: coarse features first, fine details added later. Market return sequences may not have this structure — the fine-scale noise on any given day is not a refinement of the coarse-scale trend. This may explain FinGAN's weaker performance relative to QuantGAN, whose temporal convolutions are better suited to sequential dependencies.

Why It Matters Here — link to the thesis

FinGAN serves as a baseline for CNN-based approaches to financial time series. Its architecture is deliberately simple (3 upsampling blocks, no attention, no recurrence) to isolate the effect of the upsampling inductive bias. The sq_len divisibility-by-8 constraint must be checked before training on any new dataset.

Chapter 15

Evaluation: Metrics, Taxonomy, and the Shuffled-Control Audit

15.1 The Metric Taxonomy

Situation — the everyday picture

You built a machine that generates fake playing cards. You want to check whether the fakes are good. You could count the card values (does the fake deck have the right number of aces?) — that's a *fidelity* check. But you could also shuffle the deck and see if the pattern looks natural — that's a *temporal* check. Both are necessary; neither alone is sufficient.

Formal Name — the mathematics

We classify all evaluation metrics into three groups:

Fidelity metrics (7, permutation-invariant): measure the marginal distribution of r_t only. A metric m is permutation-invariant if $m(\pi(r)) = m(r)$ for any permutation π. These metrics score the shuffled control perfectly by construction.

- `mean_diff`: $|\bar{r}_{\text{real}} - \bar{r}_{\text{syn}}|$
- `std_diff`: $|\text{sd}(r_{\text{real}}) - \text{sd}(r_{\text{syn}})|$
- `wasserstein`: $W_1(p_{\text{real}}, p_{\text{syn}})$
- `quantile_mse`: $K^{-1} \sum_k (Q_{\text{real}}(\alpha_k) - Q_{\text{syn}}(\alpha_k))^2$
- `tail_index_diff`: $|\hat{\alpha}_{\text{real}} - \hat{\alpha}_{\text{syn}}|$ (Hill estimator, top 5%)
- `extreme_events_diff`: $|P_{\text{real}}(|r| > 2\sigma) - P_{\text{syn}}(|r| > 2\sigma)|$
- `energy_distance`: $2\mathbb{E}\|X-Y\| - \mathbb{E}\|X-X'\| - \mathbb{E}\|Y-Y'\|$

Temporal metrics (7, ordering-sensitive): measure properties that depend on the sequence of r_t. The shuffled control scores poorly on all of these.

- `acf_returns_mae`: ACF-MAE of r_t at lags 1–20 (tests fact 2)
- `acf_absolute_mae`: ACF-MAE of $|r_t|$ (tests volatility clustering, fact 3)
- `acf_squared_mae`: ACF-MAE of r_t^2 (tests ARCH effects, fact 6)
- `hurst_diff`: $|H_{\text{real}} - H_{\text{syn}}|$ applied to $|r_t|$ (tests fact 4)
- `resid_kurtosis_diff`: $|\kappa_{\text{real}} - \kappa_{\text{syn}}|$ for GARCH residuals (tests fact 7)
- `discriminative_auc_dist`: $|\text{AUC} - 0.5|$ from logistic classifier
- `discriminative_auc_absz`: $|(\text{AUC} - 0.506)/0.084|$ (z-score vs empirical null)

Descriptive metrics (7, excluded from composite): computed and displayed for diagnosis only.

- `kurtosis_diff`: $\mathbb{E}[X^4] = \infty$ for $\hat{\alpha} < 4$; sample values diverge with n

- `skewness_diff`: $\mathbb{E}[X^3] = \infty$ for $\hat{\alpha} < 3$ (4/5 BRICS markets)
- `arch_pvalue_diff`: saturates on real data ($|\Delta p| = 0.000000$)
- `arch_stat_diff`: null sd 46.4 on simulated GARCH; too noisy to rank
- `var_coverage_error`: unconditional VaR; cannot distinguish Gaussian noise from real
- `garch_persistence_diff`: sd 0.31 across seeds; range 0.000–0.985
- `discriminative_auc_raw`: raw AUC value (superseded by dist/absz variants)

Why It Matters Here — link to the thesis

The three-group taxonomy is the central methodological contribution of this paper. The composite score = (fidelity_rank + temporal_rank)/2 gives equal weight to both families regardless of how many individual metrics each contains. The shuffled control is always shown in the results table with NaN ranks — it serves as the adversarial baseline.

15.2 The Shuffled-Control Audit

Situation — the everyday picture

You buy a set of playing cards supposedly shuffled randomly. To test the shuffler, you check: (1) are the right cards present (4 of each value)? (2) is the ordering random? Test (1) passes trivially — you can count cards without looking at order. Test (2) is the harder test that actually verifies the shuffle.

Worked Arithmetic — concrete numbers

Shuffled control construction:

```
rng = numpy.random.default_rng(42)
shuffled = rng.permutation(pooled_real_array).copy()
```

Scores of shuffled control on

Fidelity metrics (pooled test set):

Metric	Shuffled control
wasserstein	0.000e+00 (exact zero)
kurtosis_diff	1.78×10^{-15} (floating-point noise)
energy_distance	2.97×10^{-5}
mean_diff	$< 10^{-14}$
quantile_mse	$< 10^{-14}$
acf_absolute_mae	0.076 (correctly penalised)
hurst_diff	0.052 (correctly penalised)

Under the old unweighted 19-metric composite: avg_rank = 1.24 (shuffled) vs 2.47 (real-like generator). The shuffled deck won.

Formal Name — the mathematics

A metric m is **permutation-invariant** if for all $r \in \mathbb{R}^n$ and all permutations $\pi \in S_n$:

$$m(\pi(r_1, \ldots, r_n)) = m(r_1, \ldots, r_n)$$

Any metric that depends only on the *sorted* values (empirical CDF, moments, quantiles, order statistics) is permutation-invariant. This includes: Wasserstein, energy distance, KS statistic, kurtosis, skewness, mean, variance, tail index, extreme event rate.

A metric m is **ordering-sensitive** if $\exists\, r, \pi$ such that $m(\pi(r)) \neq m(r)$. ACF, Hurst exponent, GARCH-based metrics, and the discriminative classifier (which uses time-window features) are all ordering-sensitive.

Where the Analogy Breaks

The shuffled control is not a bad generator — it is the best possible fidelity generator (it *is* the real data). Its failure on temporal metrics is not a bug; it is the point. Any evaluation that does not include the shuffled control as a baseline cannot claim to have tested temporal structure.

Why It Matters Here — link to the thesis

The shuffled control is a permanent entry in the evaluation table (with NaN composite rank). It plays the same role as a positive control in a biology experiment: it is expected to score well on fidelity and poorly on temporal. A GAN that scores worse than the shuffled control on temporal metrics has learned nothing about market dynamics.

15.3 Discriminative AUC: Null Calibration and Direction Bug

Situation — the everyday picture

You train a classifier to tell real returns from synthetic returns. If it achieves 50% accuracy (random guessing), the synthetic data is indistinguishable from real. If it achieves 90%, the synthetic data is clearly different.

Worked Arithmetic — concrete numbers

Direction bug: under ascending rank (lower is better), AUC $= 0.30$ ranked above AUC $=$ 0.50. But AUC $= 0.50$ is *better* (indistinguishable), and AUC $= 0.30$ is worse (the classifier is *anti-predictive* — it can tell real from fake by doing the opposite of its prediction). Fix: rank on $|\mathrm{AUC} - 0.5|$.
Null calibration: even with two samples from the same distribution, the classifier achieves AUC $\neq 0.5$ exactly. Empirical null (15 real-vs-real half-splits): 0.506 ± 0.084.
An observed AUC $= 0.62$: $z = (0.62 - 0.506)/0.084 = 1.36$, $p = 0.17$. Not significant.
CV shuffle bug: 20-day windows overlap by 19 observations. Adding `shuffle=True` to the K-fold cross-validation puts near-duplicate windows in both train and test, creating look-ahead leakage. Measured null shift: $0.506 \to 0.584$ with shuffled CV. Unshuffled CV is correct.

Formal Name — the mathematics

AUC (Area Under the ROC Curve) for a binary classifier: $\mathbb{P}(\hat{p}(x_{\text{real}}) > \hat{p}(x_{\text{syn}}))$.
The optimal value for a generator is AUC $= 0.5$ (chance level).
Our two ranking metrics:

$$\texttt{discriminative\_auc\_dist} = |\mathrm{AUC} - 0.5|$$

$$\texttt{discriminative\_auc\_absz} = |(\mathrm{AUC} - 0.506)/0.084|$$

Both are minimised at the GAN optimum and entered into the Temporal family.

Why It Matters Here — link to the thesis

The three bugs found in the discriminative AUC implementation (direction, null calibration, CV shuffling) are documented in ED Table 2 of the paper. They represent common pitfalls in synthetic data evaluation that affect published benchmarks. The corrected implementation is in cell `aaef7bd2` of `3_4_integrated_pipeline.ipynb`.

Chapter 16

Downstream Utility: VaR Backtesting and QLIKE

16.1 Value at Risk

Situation — the everyday picture

A bank risk manager wants to answer: "What is the most I should expect to lose on any single trading day, with 95% confidence?" The answer is the **Value at Risk** (VaR) at the 5% level. If the daily VaR is 2%, it means the bank should expect to lose more than 2% on at most 1 day in 20.

Worked Arithmetic — concrete numbers

Unconditional VaR (5th percentile of the empirical distribution): For BOVESPA 20-year data, the 5th percentile of $r_t \approx -2.7\%$. $\mathrm{VaR}_{0.05} = -2.7\%$ means: on 5% of days, expect worse than a -2.7% return.
Conditional VaR (GARCH-based):

$$\hat{\sigma}_t^2 = \hat{\omega} + \hat{\alpha} r_{t-1}^2 + \hat{\beta}\hat{\sigma}_{t-1}^2$$
$$\mathrm{VaR}_t = \hat{\mu} + \hat{\sigma}_t \cdot z_{0.05}, \quad z_{0.05} = -1.645$$

This VaR changes each day based on yesterday's volatility. On a quiet day, $\mathrm{VaR}_t \approx -1.5\%$. After a large move, $\mathrm{VaR}_t \approx -3.5\%$.
Why unconditional VaR is useless as an evaluation metric: Gaussian iid noise (white noise with σ = historical vol) achieves coverage error = 0.0276. Real GARCH parameters on real data also achieve coverage error = 0.0259. The two are nearly identical because unconditional VaR tests only the marginal distribution, which is already covered by Wasserstein and KS.

Formal Name — the mathematics

Value at Risk at level α:

$$\mathrm{VaR}_\alpha = \inf\{x : \mathbb{P}(r_t \leq x) \geq \alpha\}$$

For a GARCH model with normal innovations:

$$\mathrm{VaR}_{\alpha,t} = \hat{\mu} + \hat{\sigma}_t \Phi^{-1}(\alpha)$$

Coverage error: $|\hat{v} - \alpha|$ where $\hat{v}$ = fraction of test days with $r_t < \text{VaR}_{\alpha,t}$. Correct model: $\hat{v} \approx \alpha = 5\%$. Observed for real GARCH on real BRICS data: $\hat{v} = 7.59\%$, so coverage error $= |0.0759 - 0.05| = 0.0259$.

Where the Analogy Breaks

VaR captures the 5th percentile only. It says nothing about what happens *beyond* the 5th percentile (Expected Shortfall / CVaR covers this). We use VaR because Kupiec and Christoffersen backtests are specifically designed for it, and because the thesis focuses on evaluating the GARCH structure, not the full tail.

Why It Matters Here — link to the thesis

The real-GARCH coverage error of 0.0259 is the reference baseline in `var_coverage_error`. A synthetic GARCH that transfers well should produce VaR forecasts with similar coverage. This metric is in DESCRIPTIVE (not TEMPORAL) because it probes only the marginal coverage rate, not the temporal dynamics of whether violations cluster together (which is what Christoffersen tests).

16.2 Kupiec Backtest

Situation — the everyday picture

If your VaR model is correct, exceedances (days where the loss exceeds VaR) should happen exactly 5% of the time. If you observe 40 exceedances in 200 test days (20%), something is wrong. The Kupiec test asks: is the observed exceedance rate statistically distinguishable from the claimed 5%?

Worked Arithmetic — concrete numbers

Let n = test days, x = observed exceedances, α = claimed coverage (5%).
Under the null hypothesis ($p = \alpha$):

$$LR_{\text{Kupiec}} = -2\ln\left[\frac{\alpha^x(1-\alpha)^{n-x}}{\hat{p}^x(1-\hat{p})^{n-x}}\right], \quad \hat{p} = x/n$$

$LR_{\text{Kupiec}} \sim \chi^2(1)$ under H_0.
Power at different n: $n = 125$ (5-year per-market): power = 17% at true $p = 7\%$ — barely above random. $n = 496$ (20-year per-market): power $\approx 56\%$. $n = 2{,}480$ (20-year pooled): power $\approx 99\%$.
With 5-year data, Kupiec is essentially powerless. With 20-year pooled data, it is meaningful.

Formal Name — the mathematics

Kupiec (1995) unconditional coverage test:

$$H_0 : p = \alpha, \quad H_1 : p \neq \alpha$$

Likelihood ratio statistic:

$$LR_{uc} = 2\ln\left[\hat{p}^x(1-\hat{p})^{n-x}\right] - 2\ln\left[\alpha^x(1-\alpha)^{n-x}\right] \sim \chi^2(1)$$

Rejects at 5% significance if $LR_{uc} > 3.84$.

Where the Analogy Breaks

Kupiec tests only the *unconditional* coverage: is the total fraction of exceedances right? It does not test whether exceedances cluster in time (which would indicate the model fails in precisely the turbulent periods that matter most). That is the Christoffersen test.

Why It Matters Here — link to the thesis

Kupiec p-values are reported in the downstream utility section but are descriptive only at per-market n. At pooled $n \approx 2{,}480$, they are informative. The primary ranking metric is QLIKE because it is a proper scoring rule with no minimum-sample-size issue.

16.3 Christoffersen Independence Test

Situation — the everyday picture

Your VaR model has 5% coverage (right number of exceedances). But all 5% happen in a single turbulent week. After that week, perfect calm. The model is correct on average but wrong when it matters: it fails during exactly the periods banks need accurate risk estimates.

Formal Name — the mathematics

Christoffersen (1998) independence test: tests whether VaR exceedances are independent over time.
Let $I_t = \mathbf{1}[r_t < \mathrm{VaR}_{\alpha,t}]$ be the indicator of an exceedance. Define transition counts: n_{ij} = number of days where $I_{t-1} = i$ and $I_t = j$.
Under independence:

$$LR_{\text{ind}} = 2 \ln \left[\frac{\hat{\pi}_{01}^{n_{01}} (1 - \hat{\pi}_{01})^{n_{00}} \hat{\pi}_{11}^{n_{11}} (1 - \hat{\pi}_{11})^{n_{10}}}{\hat{\pi}^{n_{01}+n_{11}} (1 - \hat{\pi})^{n_{00}+n_{10}}} \right] \sim \chi^2(1)$$

where $\hat{\pi}_{ij} = n_{ij}/(n_{i0} + n_{i1})$ and $\hat{\pi} = (n_{01} + n_{11})/(n_{00} + n_{01} + n_{10} + n_{11})$.

Why It Matters Here — link to the thesis

The combined Christoffersen test ($LR_{\text{cc}} = LR_{\text{uc}} + LR_{\text{ind}} \sim \chi^2(2)$) tests both unconditional coverage and independence simultaneously. For synthetic GARCH parameters to be useful, the VaR forecasts they produce must have both correct coverage and independent exceedances. This is a substantially harder bar than Kupiec alone.

16.4 QLIKE Loss

Situation — the everyday picture

You have two weather forecasters. Forecaster A says "tomorrow will be a 2% return day" (one number). Forecaster B says "tomorrow's volatility will be 1.5%" (a distribution). You observe the actual return. How do you score Forecaster B, who gave you a distribution, not a point forecast?

Worked Arithmetic — concrete numbers

QLIKE (Patton 2011):

$$\text{QLIKE}_t = \log \hat{\sigma}_t^2 + r_t^2/\hat{\sigma}_t^2$$

A volatility forecast $\hat{\sigma}_t$ that is too high penalises $\log \hat{\sigma}_t^2$ (unnecessarily large). A forecast too low penalises $r_t^2/\hat{\sigma}_t^2$ (missed a large move).
TRTR (train and test on real data) baseline: QLIKE ≈ -8.134, sd 0.000. A synthetic GARCH that produces parameters close to the true ones will have QLIKE close to -8.134.
Per-market inversion (5-year data, $n \approx 126$): Gaussian noise: QLIKE $= -6.888$. Real data: QLIKE $= -6.876$. Gaussian noise beats real data — QLIKE is meaningless at this n. Shuffled control sd across seeds: 3.354 — random variation dominates.
Pooled (20-year, $n \approx 2{,}480$): QLIKE is stable and interpretable. Per-market ($n \approx 496$) also stable.

Formal Name — the mathematics

QLIKE (Quasi-Likelihood): a proper scoring rule for volatility forecasts.

$$\text{QLIKE}(r, \hat{\sigma}) = \frac{1}{T}\sum_{t=1}^{T}\left[\log \hat{\sigma}_t^2 + \frac{r_t^2}{\hat{\sigma}_t^2}\right]$$

A scoring rule is **proper** if it is minimised in expectation when $\hat{\sigma}_t^2 = \mathbb{E}[r_t^2|\mathcal{F}_{t-1}]$ (the true conditional variance). Patton (2011) shows QLIKE is robust to the use of imperfect variance proxies (here: r_t^2 instead of the true σ_t^2).

Where the Analogy Breaks

QLIKE is minimised (more negative = better) when the volatility forecast matches the conditional variance. It cannot test whether the GARCH model is the right *specification* (e.g., whether leverage effects matter), only whether the estimated parameters are close to the true conditional variance. A synthetic GARCH with wrong parameters but right unconditional volatility might score well on QLIKE by accident — Kupiec and Christoffersen provide additional checks.

Why It Matters Here — link to the thesis

QLIKE is the primary downstream utility ranking signal because: (1) it is a proper scoring rule, (2) it directly tests whether the GARCH structure learned from synthetic data transfers to real data, and (3) it is continuous with no minimum-power issues at pooled $n \approx 2{,}480$. The TSTR protocol (Train on Synthetic, Test on Real) uses synthetic GARCH parameters to forecast real-data volatility and computes QLIKE on the real test set.

16.5 Walk-Forward Validation

Situation — the everyday picture

You want to evaluate a forecasting model on historical data. The naive approach: train on all data up to 2020, test on 2020–2026. But this uses only one test period. If 2022–2026 happened to be unusual (it was, for MOEX), your evaluation might be misleading.
Walk-forward validation uses multiple test periods, each with a fresh training set.

Formal Name — the mathematics

Rolling-origin walk-forward (CTBench protocol):

- Divide data into K folds (here: $K = 5$)
- For fold k: train on data up to fold k's start, test on fold k
- Report metric mean $\pm$ std across 5 folds

Unlike K-fold cross-validation, the order of folds is never shuffled. Later folds always use more training data (expanding window) or a shifted training window (rolling window). We use rolling origin.

Why not K-fold with shuffling? Financial returns have serial dependence. Shuffling folds puts future data in the training set (look-ahead leakage). The discriminative AUC case showed this explicitly: shuffling 20-day windows shifts the null from 0.506 to 0.584.

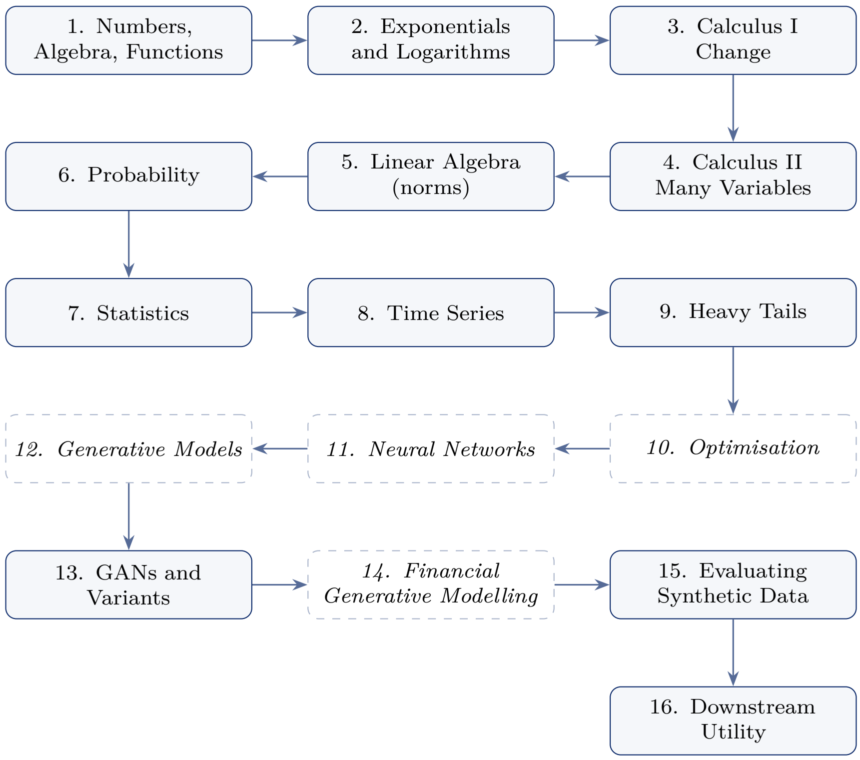
Why It Matters Here — link to the thesis

Five rolling folds per market give mean $\pm$ std across folds rather than a single metric estimate. This enables: (1) confidence intervals, (2) detection of fold-specific anomalies (MOEX fold 3 covers the 2022 invasion), (3) comparison of temporal stability across architectures. The CTBench protocol (Liao et al., 2025) requires this design for standardised reporting.

Symbol Glossary

Symbol	Meaning
r_t	Log return on day t: $\ln(P_t/P_{t-1})$
σ_t	Time-varying conditional standard deviation at t
σ	Unconditional (long-run) daily volatility
μ	Long-run mean return (close to zero)
α (GARCH)	ARCH coefficient: weight on the squared previous shock
β (GARCH)	GARCH coefficient: weight on previous variance
ω	GARCH intercept: ensures long-run variance is positive
$\hat{\alpha}$ (Hill)	Tail index estimate from Hill estimator
H	Hurst exponent ($H > 0.5$ implies long memory)
W_1	Wasserstein-1 (Earth Mover's) distance
$\mathbb{E}$	Expectation operator
Var	Variance operator
$\mathcal{N}(\mu, \sigma^2)$	Normal distribution with mean μ and variance σ^2
$\mathbb{P}(\cdot)$	Probability of an event
z_α	α-quantile of standard normal: $z_{0.05} = -1.645$
AUC	Area under the ROC curve
GAN	Generative Adversarial Network
WGAN-GP	Wasserstein GAN with Gradient Penalty
GRU	Gated Recurrent Unit
TCN	Temporal Convolutional Network
CNN	Convolutional Neural Network
BCE	Binary Cross-Entropy loss
GARCH	Generalised ARCH
VaR	Value at Risk
QLIKE	Quasi-likelihood loss for volatility forecast evaluation

Dependency graph and roadmap. Chapters are strictly ordered: nothing depends on a concept introduced later. The foundations (Part I) are prerequisites for everything that follows.



Solid boxes are written. Dashed italic boxes are planned and not yet drafted; their numbers are reserved so that cross-references remain stable as the book is completed. The critical path this book exists to support runs

$$\text{derivative} \to \text{partial derivative} \to \text{gradient} \to \text{norm} \to \text{Lipschitz continuity} \to \text{Wasserstein distance} \to \text{WGAN} \to \text{critic} \to \text{gradient penalty},$$

and Chapters 3–5 supply every link in it up to and including the norm.